



WEST VISAYAS STATE UNIVERSITY
CALINOG CAMPUS

CAPSTONE PROJECT

H A N D B O O K

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Capstone Project Handbook

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ISBN:

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Published by:

Lay Out/Cover Design : Kent Alvin M. Cebal

PREFACE

This book has been the result of the author's many years of experience teaching thesis writing and capstone project. From her work as a teacher in the thesis writing and capstone project, she learned exactly the kind of knowledge and skills needed by the students who are taking the course and eventually will write their research reports and develop their projects. Therefore, she tried as much as she could to incorporate in this book, discussions, and explanations of all the essential ideas, concepts, and principles in methods of research in computing and the skills needed in thesis writing and developing of a capstone project.

No claim is made to originality. Those familiar with writings about methods of research, thesis writing, and capstone project will notice that many of the important ideas, concepts, and principles found in this book came from different sources with only some sprinklings of original ideas. The author is, therefore, expressing her thanks and gratitude to all those authors whose ideas, concepts, and principles have been used to enrich the content of this book. It should be noted here that they are fully recognized and acknowledged and they have always been cited for every bit of idea or information borrowed from them. They are cited in the different pages, at the end of every unit and are also listed in the references.

The author also acknowledges her debt of gratitude to all those who in one way or another, helped in the making of this book. c/o Sir Jovas

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UNIT-I: REVIEW OF INTRODUCTION TO CAPSTONE PROJECT

INTENDED LEARNING OUTCOMES (ILOs)

At the end of this Unit; the students must have:

1. defined research and its relation to goals and purpose;
2. discussed purpose, methods, and characteristics of research;
3. differentiated research to design;
4. differentiated thesis against a capstone project; and
5. discussed CHED's suggested documentation template/format

1.1 Research

As Seliger and Shohamy 1989 said, research is “the formalization of natural processes we all carry from birth in dealing with the environment.” It begins with curiosity about a certain phenomenon and it attempts to seek the truth or facts about the relationships of the variables that influence the observed phenomenon. It aims to ‘describe, identify, control relationships among phenomena’ to test existing principles and develop a theory.

Research may be defined as a scientific investigation of phenomena which includes collection, presentation, analysis, and interpretation of facts that links man's speculation with reality.

“...the systematic process of collecting and analyzing information (data) in order to increase our understanding of the phenomenon about which we are concerned or interested”.

More Understanding on what is Research?

- Research is a careful and systematic process of inquiry to find answers to problem of interest.
- Research include: use of facts, use of theories, data (fact) analysis, sampling, doing an experiment, going to library to read up on a topic
- To do ‘research’ is to investigate the problem systematically and thoroughly
- Goal of research is to solve ‘problem’ of interest. Is simply a systematic and refined technique of thinking, employing specialized tools, instruments, and procedures in order to obtain a more adequate solution to a problem than would be possible under ordinary means (Crawford, 1967).

1.2 Purpose of Research

We humans are great knowledge accumulators. We love knowing about everything. By conducting research, we attempt to get closer to reality by attempting to develop or construct new knowledge.

The purpose of research is to enhance society by advancing knowledge through the development of scientific theories, concepts and ideas.

The purpose of research is to serve man and the goal of research is good life, to satisfy man's craving for more understanding, to improve his judgments, to add to his power, to reduce the burden of work, to relieve suffering, and to increase the satisfaction in multitudinous ways- these are the large and fundamental goals of research (Good, 1972).

For specific purposes and goals of research, the following may be mentioned:

- a. Researches carry out a scientific source for any practice or methodology in any field or discipline. It is important because the interest of whole system or operation and all the concerned members of the institution or organization is at stake.
- b. Research is necessary for continuous development of and further productivity in any field like education, management, business, engineering and other areas of endeavor.
- c. Research develop tools for assessing the effectiveness of any practice and operation.
- d. Research provides solutions to problem concerning almost all issues encountered in different areas of work.
- e. Research develops and evaluates alternative approaches to t educational aspects of any discipline that enable the students to gain broad knowledge and specialized skills for safe practice.
- f. Research advances the personal and professional qualifications of a practitioner.

1.3 Characteristics of Research

The following are the generally accepted characteristics of research (Calmorin, 2001):

- a. **Empirical** – Research is based on direct experience or observation by the researcher. The collection of data relies on practical experience without benefit of the scientific knowledge or theory.
- b. **Logical** – Research is based on valid procedures and principles. Scientific investigation is done in an orderly manner so that the researcher has confidence on the results. Logical examination of the procedures used in the research enables the researcher to draw valid conclusions. Thus, the logic of valid research makes it important for decision making.
- c. **Cyclical** – Research is a cyclical process. It starts with a problem and ends with a problem. For instance, a researcher who completes his study states his findings and draws up his conclusions and recommendations. In his recommendations, many problems may crop up as other subjects for study hence, the cycle is repeated.
- d. **Analytical** – Research utilizes proven analytical procedures in gathering the data, whether historical, descriptive, experimental or case study.
- e. **Replicability** – The research designs and procedures are replicated to enable the researcher to arrive at valid and conclusive results. Similarities and differences of replicated researchers can be compared. The more replications, the more valid and conclusive the results would be.

- f. **Critical** – Research exhibits careful and precise judgments. A higher level of confidence must be established, i.e., at one percent or five percent level of confidence.

1.4 Research Methods

A research methodology defines what the activity of research is, how to proceed, how to measure progress, and what constitutes success (Aquino, 1974);

- a. **Descriptive Research** – it describes what is. It involves the description, recording, analysis, and interpretation of the present nature, composition or processes of phenomena. The focus is on prevailing condition.
- b. **Historical Research** – a process of selecting the area or topic to write the history about, collecting data about events that occurred in the area or about the topic, collating the data, sifting the authentic from non-authentic, and then making an interpretative narrative about or critical inquiry into the whole truth of the events. Historical research describes what occurred in the past and makes a critical inquiry into the truth of what occurred.
- c. **Experimental Research** - a problem-solving approach that describes the effect when certain variables are carefully controlled or manipulated. It focuses on the past, present and future conditions.
- d. **Ex post Facto Research** – in this method, the researcher investigates a problem by studying the variables in retrospect. It is a research in which the dependent variable is immediately observable and the main concern is to find out the antecedents that gave rise to this consequence (Sevilla, 1984).
- e. **Case Study Research** – according to Bromley (1990), it is a systematic inquiry into an event or a set of related events which aims to describe and explain the phenomenon of interest.

1.5 Design

In the area of engineering design is the process of devising a system, component, or process to meet desired needs. It is a decision-making process (often iterative), in which the basic sciences, mathematics, and the engineering sciences are applied to convert resources optimally to meet these stated needs (ABET, 2015).

The keyword in design is the word “optimum”. Good designs convert and/or uses IT resources optimally to meet the stated needs. It is for this reason why the Commission on Higher Education (CHED) defines the Bachelor Science in Information Technology (BSIT) program as the study of the utilization of both hardware and software technologies in planning, installing, customizing, operating, managing, administering and maintaining IT infrastructure that provides computing solutions to address the needs of an organization. Considering that the term Research and Design are two different terms, Bachelor of Science in Computer Science (BSCS) program is a research based-program thesis is a requirement. The term design, on the other hand is the applicable term for BSIT, Bachelor of Science In Information

System (BSIS), and Bachelor of Science in Entertainment and Multimedia Computing (BSEMC) programs because these programs require capstone project (Lavinia, 2016).

Research and Design Distinguished

Ashley Karr (2013) distinguished research and design in this wise, thus:

To research something is to investigate it systematically. We do this in order to reach new conclusions, establish new facts, and learn as much about the truth as possible. Research also gives us a chance to find problems that we can potentially fix. We research to understand the world.

To design something is to create the form and function of an object, system, or instruction. We do this in order to make out experience here on earth better, safer, healthier, more comfortable, more fun –and we can create solutions to the problems we found during the research. We design to change the world. Simply stated, we research to explain WHY THINGS happen; we design in order TO MAKE THINGS HAPPEN.”

The American Society for Engineering Education (2006) explains the difference between research and design;

Design is NOT research, which may be defined as “ a careful investigation or study, especially of a scholarly or scientific nature. A design task may require research to accomplish a task, but it typically involves the integration of knowledge, not the creation of knowledge.” Design is different than research. Design considers alternative solutions by selecting the optimal solution with a fixed goal or specifications in mind. Design often results in a commercial product being developed. Research has an open-ended goal and is exploratory, with no set specifications in mind, and does not necessarily result in a product or a service.

Table 1. Comparison of research and design

Research	Design
It must be new.	It does NOT need to be new.
It does not have to be good.	It has to be good.
In sum, research does not have to be good but it must be new.	In sum, design does not have to be new, but it must be good.
It is done to understand the world.	It is done to change the world.

1.6 What is a Thesis?

A thesis may be defined as the report of a scholar upon some piece of research which he has completed. It is the culmination of a devious process extending from the initial insight into the opportunity for investigation to the insertion of the final footnote (Cole and Bigelow, 1935). In the area of computing, it is defined as a technical report on a systematic investigation of a problem that can be solved using Computing. It may include a solution, an approximate or partial solution, and a scientific investigation, or the development of results leading to the solution of the problem (Section 2.1, Article II of Annex A of CMO 25 s. 2015).

1.7 What is a Capstone Project?

A Capstone Project is an undertaking appropriate to a professional field. It should significantly address an existing problem or need. An information Systems (IS) Capstone Project focuses on business processes and the implications of introducing a computing solution to a problem while an Information Technology (IT) Capstone Project focuses on the infrastructure, application, or processes involved in implementing Computing solution to a problem (Section 2.2, Article II of Annex A of CMO 25 s. 2015). Of course, for the BSEMC program the capstone project deals with game development and digital animation Technology (Section 8.2.3 of CMO 2 series of 2014)

Under OBE parlance, a capstone project is the design completed by a student, which results in an optimum use of the resources demonstrating his competencies gained from his educational experiences.

1.8 CHED'S Suggested Documentation Template/Format

The following are the sample thesis and capstone project format based on Annex A of CMO 25 series of 2015. The authors would like to emphasize that there is NO single correct thesis template or format. The school, in the exercise of its academic freedom, may use other format(s).

Table 2. CHED'S Suggested Documentation Templates/Format

CS Thesis (Foundations of Computer Science)	CS Thesis (Software Development)	IT/IS Capstone Project
Title page Abstract Table of Contents List of Figures/ List of Tables, List of Notations Introduction Background of the Problem Statement of the Problem Objectives of the Study Significance Scope and Limitations	Title page Abstract or Executive Summary Table of Contents List of Figures/ List of Tables, List of Notations Introduction Project Content Purpose and Description Objectives Scope and Limitations	Title page Executive Summary Table of Contents List of Figures/ List of Tables, List of Notations Introduction Project Content Purpose and Description Objectives Scope and Limitations
Related Literatures Theoretical	Related Literatures Technical Background	Review of Related Literature/Systems
Proposed Solution to the Problem	Design and Methodology	Methodology, Results and Discussion Requirement Analysis Requirements Documentation Design of Software, Systems, Product and/or Processes Development and Testing, where applicable Description of the Prototype, where applicable Implementation Plan (Infrastructure/Deployment) where needed Implementation results, where Applicable

Results and Discussion	Results and Discussion	
Conclusions and Recommendation	Conclusions and Recommendation	Recommendations
Appendices	Appendices	Appendices

Name:	Date:
--------------	--------------

● Review Questions:

1. What is the difference between research and design?

[illegible]

2. What is the difference between Thesis and Capstone Project?

[illegible]

3. Why a research method is different from methodology?

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● Assessment Task

Answer the following situation:

Mr. De Leon is an Assistant Professor I of West Visayas State University-Calinog Campus under the School of Information and Communication Technology (SICT). He is a former professor at a private university and been teaching System Analysis and Design course for 14 years. He is curious if the research grade of the graduates from K-12 program has something to do with their skills in System Analysis and Design. So, in his class in System Analysis and Design, he asked his students about their final grade in Research. At the end of the semester, he correlated the Research grade of his students with their final grades in System Analysis and Design.

Did Mr. De Leon conduct a research or a design? Explain your answer.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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UNIT-II: WRITING THE COMPONENTS OF CHAPTER 1

INTENDED LEARNING OUTCOMES (ILOs)

At the end of this unit; the students must have:

- written a good introduction/background of the study;
- formulated the statement of the problem and objectives of the study;
- identified or draw the conceptual framework of the study;
- defined the terms used in the study;
- demonstrated the purpose and description;
- explained and justify the limitations of the study; and
- written a good Chapter 1 of the Capstone Project.

2.1 Introduction

2.1.1 Writing a Good Introduction

Laviña (2016) guidelines and example in writing a good introduction. A good introduction of an IT and IS capstone projects should contain a discussion of all the following:

1. **Presentation of the importance of the system software, algorithms, processes, or an enterprise resource plan to be designed and/or developed.** This must be the start of an introduction. In a capstone project entitled: “Development and Evaluation of a Decision Support System for the JIL Church – Calamba City Using Transaction Processing as Percieved by its Administrators and Workers for the Year 2015” the main objective of which is a development of Information System (IS) and/or a Decision Support System (DSS), part of the introduction reads in part.

Example:

Organizations have long recognized the importance of managing key resources such as labor and raw material. Information has now moved to its rightful place as a key resource. Decision makers are beginning to understand that information is not just a by-product of conducting business, rather, it fuels business and can be the critical factor in determining the success or failure of a business.

To maximize the usefulness of information, a business must manage it correctly, just as it manages other resources. Managers need to understand that costs are associated with the production, distribution, security, storage, and retrieval of information. The strategic use of information for positioning a business competitively should not be taken for granted.

2. **Research Locale.** Tell something about the organization or group where the research will be conducted. This usually the target beneficiary of the proposed system software. In the sample capstone project, the organization under study is the JIL Church at Calamba City.

Example:

It was on June 15, 1986 when the first worship service of the Jesus Is Lord Fellowship Calamba Chapter, now Jesus Is Lord Church Calamba Chapter, now Jesus Is Lord Church Calamba Chapter was held with hundreds of attendees, guests and visitors and even pastors from other Evangelical churches. But its second worship service revealed the actual number of attendees consisting of only fifty (50) estimated attendees, a resident pastor and three volunteered workers.

During the JIL's first few years of operation, information managing is not a problem. Almost all of its attendees know each other, and the resident pastor can easily recognized and call them by their names. Thus, if one attendee was not able to attend Sunday services for several times, the Follow-up Ministry can easily recognize them and make an appropriate decision and action like house visitation to discern the reasons why their brethren was unable to attend their worship service. This will enable the Follow-up Ministry to make a sound biblical advice.

3. **The existence of an unsatisfactory condition; a felt problem that needs a solution. A solution that involves development of system software.**

Example:

At present, this situation does not hold true anymore. The JIL Church now has an average of one thousand attendees per week. Sometimes, during special worship service, the total number of attendees reached one thousand five hundred at the most. And sadly, because of poor information management, the leadership was unable to produce the list of additional five hundred attendees. This made the Follow-up Ministry helpless in achieving the JIL Calamba's Vision of "Harvest 3000 plus souls." Additionally, a regular attendee who was absent during the last two to three weeks was hardly noticed. Only those popular attendees can be recognized absent and consistently they are the only members of the Church who are being followed up or being taken care of. Decision making, consequently is done under uncertainty.

4. **A desire to find a better way of doing something or improving the present system.**

Example:

It is for this reason why, the proponent, being a member of the JIL Church, gains a deepest desire to find a better way of improving the present condition by developing a software called as "Church's Decision Support System" (CDSS), that will help solve the unsatisfactory monitoring of the attendance of JIL Church members.

*If all elements of a good introduction are present so that in the end, it will be able to convince the reader or a person who shall be benefitted by the proposed project that there is really a need in doing the same, then such introduction had set the "**Project Context**" of a capstone project. Bear in mind that the "project context" forms part of the **Introduction**.*

The project context is the environment in which a project is undertaken. It has two key perspectives: internal as defined by the organizational structure, management, behavior and culture external as defined by the business environment in which the organization operates, including legislation and regulation, professional and industrial standards, and market sectors. Project sponsors and managers need to understand both the internal and external factors that can affect their projects. It

describes approaches for analyzing internal and external factors and managing stakeholders.

2.2 Background of the Study

This section presents specific information with regards to the nature of your study or system. If the system being conducted is about home security, provide a description of the process or activity, including relevant explanation and a brief but concise explanation of the home security system.

Example: (Source: Agham, R. et.al. Phil Pres Trevia, College of Technology, ISCOF-Dumangas Campus, Unpublished Undergraduate Research, 2015)

Background of the Study

Technology is already part of everyone lives. Each people are enjoying its efficiency. The development of mobile games is not just for entertainment, but it is very useful when it comes on education. Games will always be one of the ways to challenge and have fun. Making used with its evolution, the researchers are inspired to develop an educational mobile game application that focuses on Philippine Presidents.

Reminiscing the history, Philippines have so much to tell. Doing such thing will help you to identify the difference of past and present. Philippine history was one of the topics that being discuss in school. One of its highlights was the trademark made by the Philippine Presidents. This topic will always be a part of Araling Panlipunan and other subjects about the history of the Philippine Presidents. Despite of that, the Grade VI pupils of Dumangas Central Elementary School were not able to identify their images, quotes, slogan and their biography. Everyone had been part of this subject but only few could still remember their names, looks and even those rights that they have fight just to have this country a democracy.

This mobile game application would serve as learning and teaching tools and also it would give an advantage for those who have not yet encounter this kind of topic. It is also one way to refresh the minds of everyone and help them to know the things or information that they are not familiar with. It would help the pupils to improve and widen their knowledge in Philippine History especially, to the topic about leaders of the Republic of the Philippines. One of its objectives aimed to test the learning skills of a pupil to gauge their knowledge through guessing the Presidents images, quotes, slogan and knowing the Philippine Presidents biography. Based on conducted interview, the researchers have given the chance to widen their ideas. The technique used of a teacher is given an assignment before introducing the entire topic to the class.

2.3 Statement of the Problem

There should be a general statement of the whole problem followed by the specific questions or sub-problems into which the general problem is broken up.

According to (Laviña, 2016) the guidelines in formulating the general problem and the specific questions:

1. The general statement of the problem and the specific problems should be formulated first before developing the software programs.

2. Each specific questions must contribute to the development of the whole capstone project.
3. Answers to specific questions must contribute to the development of the whole capstone project.
4. The number of specific questions should be enough to cover the problem or case under study.
5. Generally, there should be a general statement of the problem and then broken up into as many specific questions as necessary.

Example:

The main problem of the project is to assist the Tabako University in the admission, registration, monitoring and advising of their students. Specifically, it attempted to answer the following questions:

1. *How to design and develop a Web-based system called Admission, Registration and Monitoring System (ARMIS) which will be capable of:*
 - a. *Providing and efficient on-line enrollment system wherein payment may be made in campus and out campus;*
 - b. *Providing an advising system by generating appropriate courses which should be registered by the students considering the pre-requisites and co-requisites of courses provided in the student's program of study;*
 - c. *Allowing faculty members to encode the grades during a period to be determined by the use;*
 - xxx
 - xxx
 - xxx
 - d. *What is the level of conformity of the ARMIS software with regard to ISO 9126 standards?*
 - e. *How to implement and deploy the ARMIS software in order to maximize organizational benefits?*

Example 2:

(Source: Orbino,B., et.al. Agri-Hayop-Alalay (AHA) Mobile Application, SICT, WVSU-CC, Unpublished Research, 2019)

Statement of the Problem

Before the objectives of the study were formulated, the problems to be solved by the system were identified. Based on the preliminary investigation with the local officers of the Municipal Agriculture Office of Calinog, the proponents formulated some problems that arose in the field of livestock farming in Calinog. This section presents the problems that occur as far as livestock farming in Calinog is concerned.

General Problem: the unavailability of quick response to the health problems of livestock animals.

Specific Problem:

1. *Lengthy time needed in treating the health issues of livestock animals.*
2. *Lack of immediate response from technicians for treating the livestock animals.*
3. *Lack of information for the farmers to use for their bases of nurturing animals.*

- 4. Limited access for sharing of information to other livestock farmers for relevant information that may be useful for their livestock animals.*

2.4 Objectives of the Study

Objectives of the Study according to (Laviña, 2016) guide the researcher and the reader to know what the proponents(s) need to do. These usually follow the chronological sequence of the statement of the problem and its specific questions. Objectives are written by either beginning with an action verb or an infinitive. For example:

- to determine the efficiency of the system as perceived by the employees; or
- identify possible problems encountered with the present system.

Example:

(Source: Laviña, C. G. , et. Al., Outcomes-Based Practical Guide to Thesis and Capstone Project Writing in Computing, Mindshapers Co., Inc. 2016)

The main objective of the project is to assist the Tabako University in the admission, registration, monitoring and advising of their students. Specifically it aims to:

- 1. Design and develop a Web-based system called Admission Registration and Monitoring System (ARMIS) which will be capable of:*
 - a. providing and efficient on-line enrollment system wherein payment may be made in campus and out campus;*
 - b. providing an advising system by generating appropriate courses which should be registered by the students considering the pre-requisites and co-requisites of courses provided in the student's program of study;*
 - c. Allowing faculty members to encode the grades during a period to be determined by the use;*

xxx
xxx
xxx

- 2. Evaluate the ARMIS in order to determine if it complies with the ISO 9126 standards;*
- 3. To prepare an implementation plan for the deployment of the ARMIS.*

xxx
xxx

Example 2 :

(Source: Orbino,B., et.al. Agri-Hayop-Alalay (AHA) Mobile Application, SICT, WVSU-CC, Unpublished Research, 2019)

Objectives of the Study

The general objective of this system: to develop a portable and mobile application that can provide solution for animal health problems.

Specific Objectives:

1. *To have quick response for animal health issues with the use of mobile application;*
2. *To provide treatment guides for common animal diseases with the use of mobile application;*
3. *To have one information center for livestock animals using a mobile application; and*
4. *To give communication solution and a group discussion about animal diseases through the use of online chatroom.*

Notice that the Objectives of the Study is the same as the Statement of the Problem. The only difference is that the latter is written in an interrogative form while the former is in the declarative form. Thus, a researcher may use either one of the two. In technical research like Computing, CHED's template strongly suggested that we use Objectives of the Study instead of Statement of the Problem for IT/IS.

2.5 Conceptual Framework

According to (Laviña, 2016) conceptual framework presents the relationship between the specific concepts that may be studied. The concepts used at this level should be derived from the concepts used in the theoretical framework. They should be more concrete.

After initial exploration of research so far on the topic, the author can make a more specific conceptual framework. It is at this point important that the author operationalizes (i.e. defines at least, and possibly indicates how concepts can be measured or judged) the various concepts in the research question method and how he/she will investigate the topic.

The conceptual framework is a collection of interrelated concepts, like a theory but not necessarily so well worked out. A conceptual framework guides your research, determining what things you will measure, and what statistical relationships you will look for.

The conceptual framework may be accompanied by a symbolic representation of a phenomenon, or a diagram usually given a schematic representation using arrows and boxes.

Example:

(Source: Laviña, C. G. , et. Al., Outcomes-Based Practical Guide to Thesis and Capstone Project Writing in Computing, Mindshapers Co., Inc. 2016)

Conceptual Framework

*The basic working knowledge of the computer necessary to function in today's world is called **computer literacy**. This knowledge includes an understanding of computer terminology, recognition of the strengths and weaknesses of the computer, an ability to use the computer although not necessarily to program it, and so on. In addition to understanding the computer, the modern manager should have information literacy. Information Literacy consists of understanding how to use information at each step of the problem-solving process, where that information can be obtained, and how to share information with others.*

Information literacy is not dependent on computer literacy. A person can be information literate but not computer literate. Ideally, however, a manager should be both computer literate and information literate.

Cisler (2000) argues there is not a binary division between information "haves and "have-nots", but rather a graduation based on different degrees of access to information technology. Compare, for example, a professor at UCLA with a high-speed connection in her office, a student in Seoul who occasionally uses a cyber-café,

and a rural activist in Indonesia who has no computer or phone line but whose colleagues in her women's group download and print out information for her. This example illustrates just three degrees of possible access a person can have to online material and illustrates the concept of digital divide.

The Study's Conceptual Framework is shown in Figure 2.

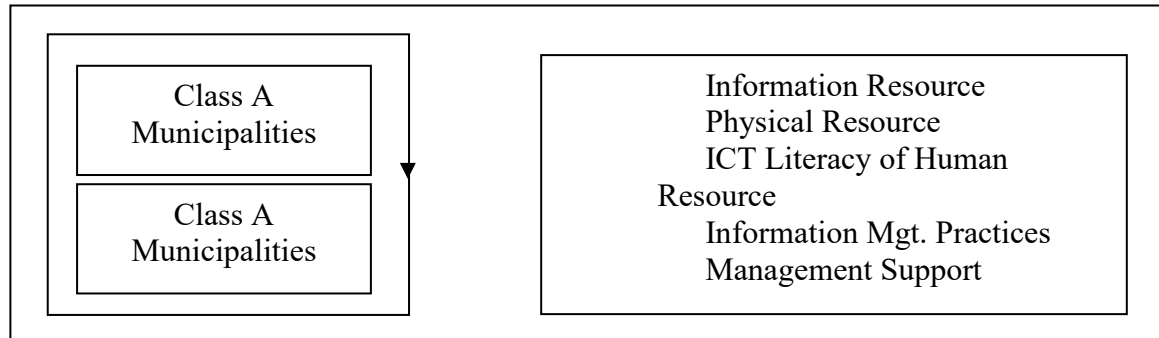
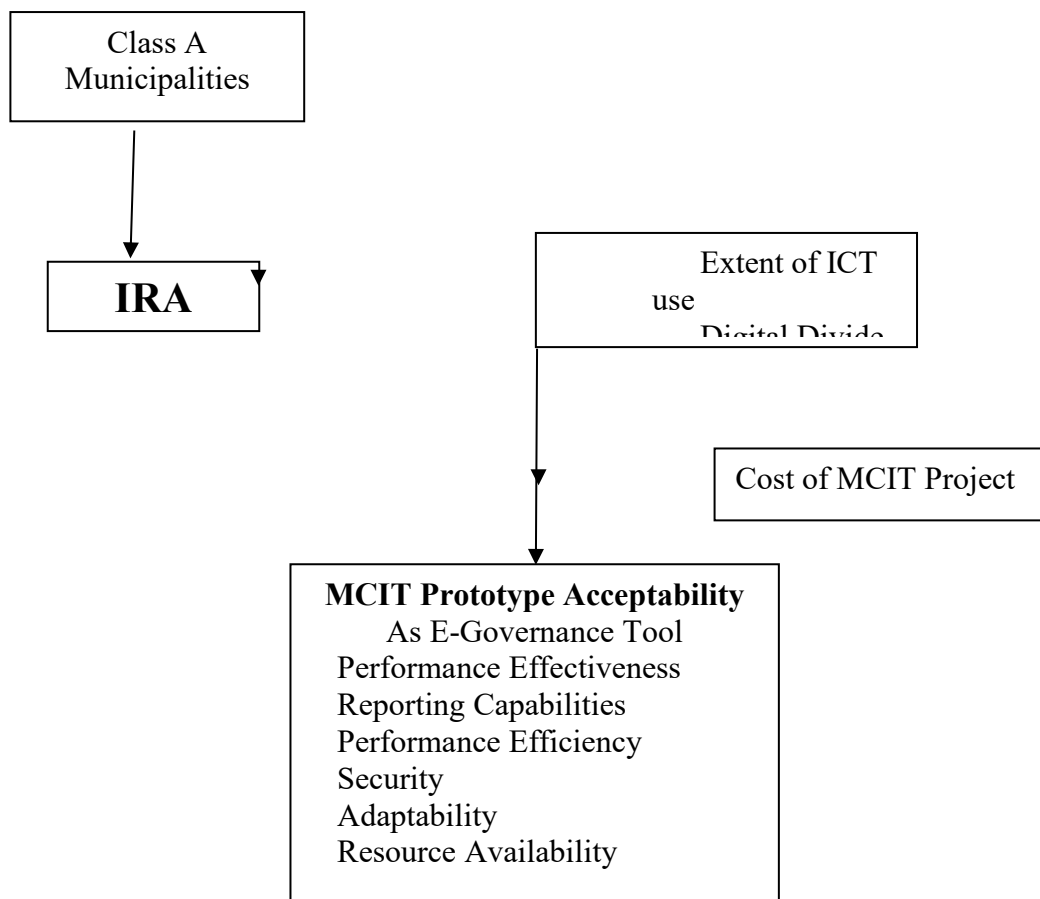


Figure 2. Conceptual Framework



Conceptual Framework of the Study

The figure depicts that the level of municipalities, which stand for the financial status of every municipality, is one factor that may influence the social resources (IRA)

of barangays, level of ICT resources, management support and ICT literacy of barangay officials and which in effect, influence the extent of ICT usage.

The differentiation of the level of the extent of ICT usage signifies digital divide among communities from the different level of municipalities.

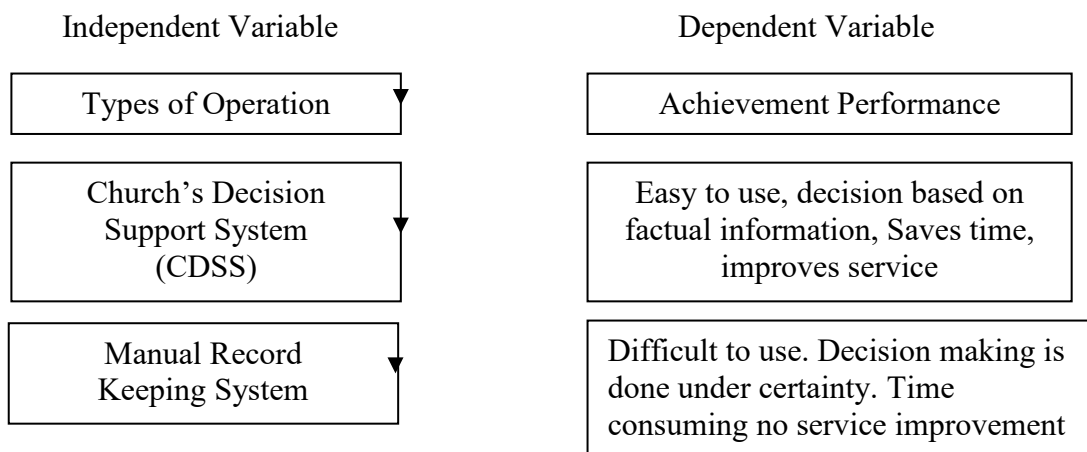
The extent of ICT usage will determine the possibility of adapting the MCIT prototype for E-Governance which will give effective information management and better service delivery for barangay constituents.

Other Styles of Developing Conceptual Framework

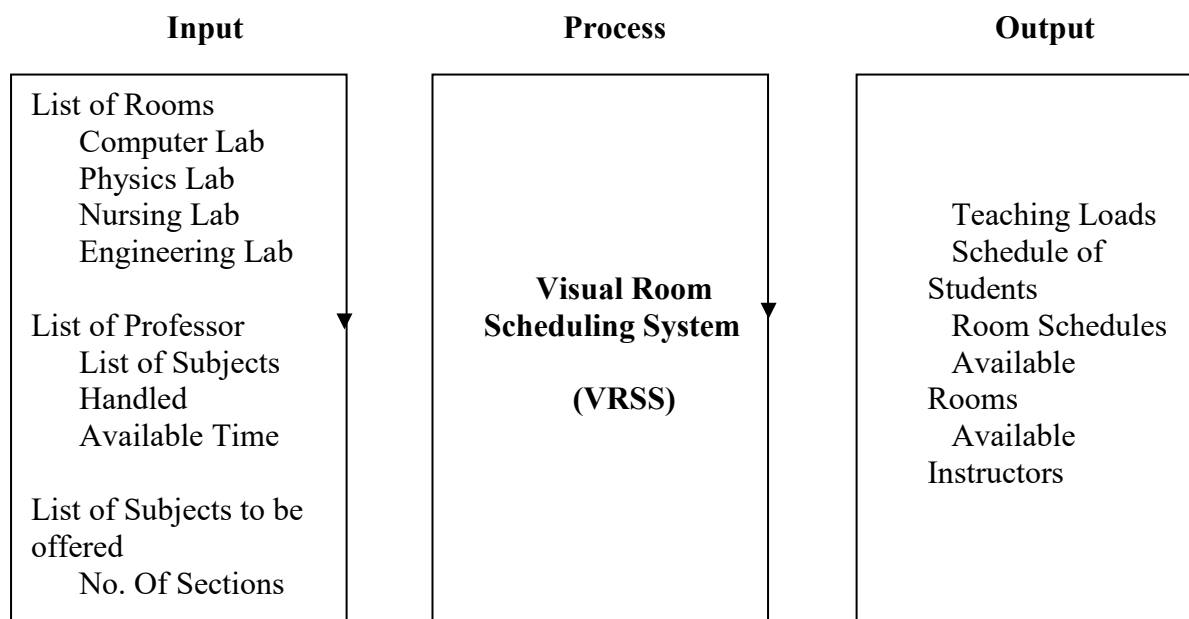
These other styles are the ones appropriate for Computing programs.

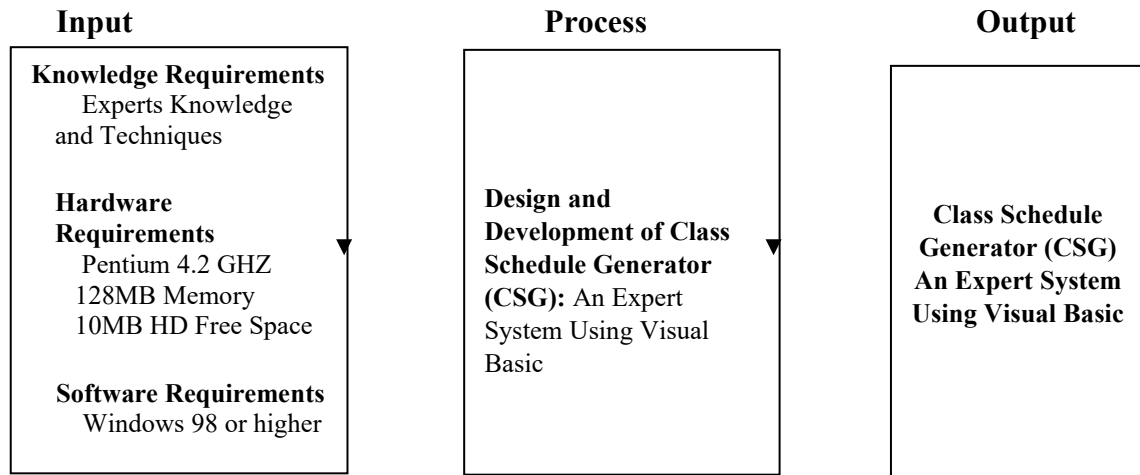
Example 1. (Usually applied in social research. This is NOT usually applied in computing researches.)

Paradigm of the Independent and Dependent Variable



Example 2. (Uses the Input Process Output (IPO) chart. This is what we usually do in technical research.)

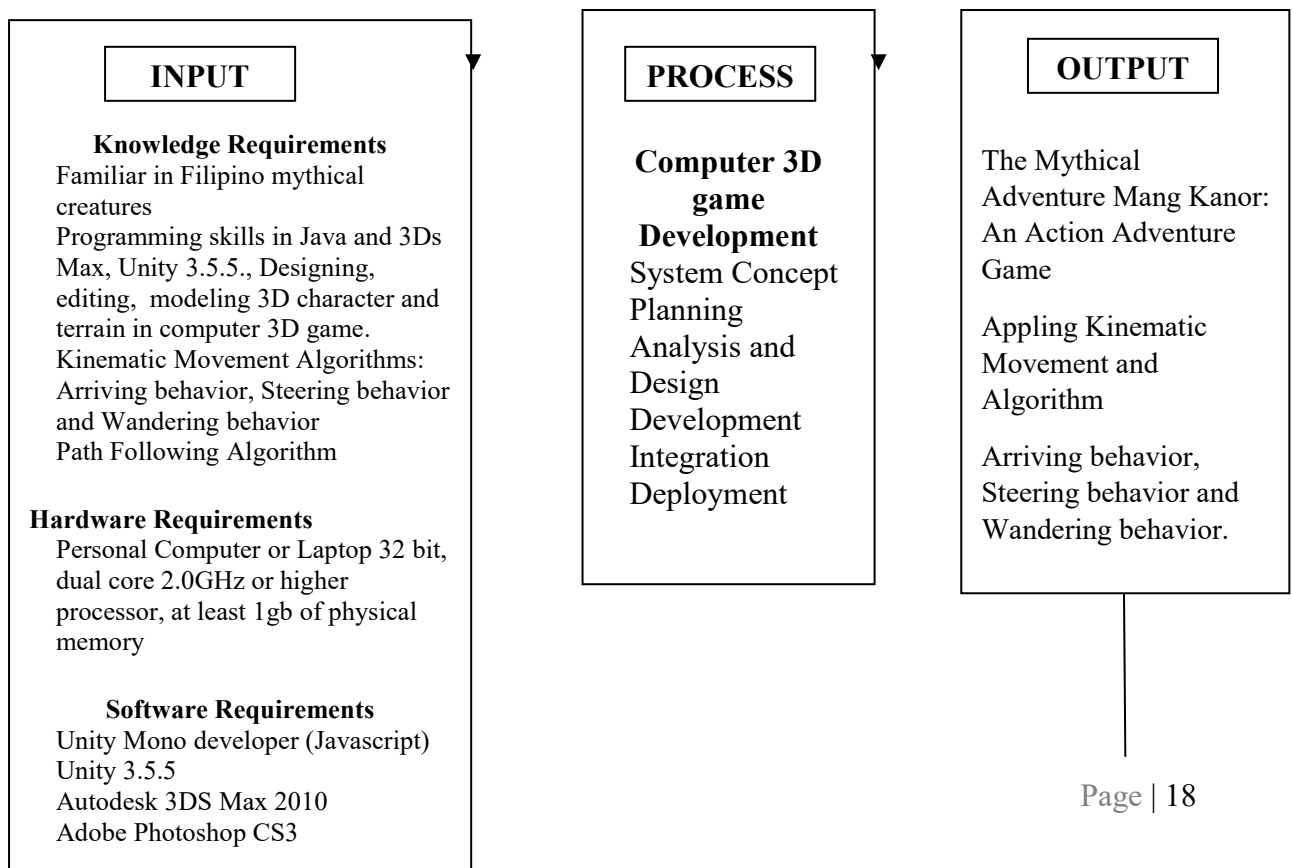




Notice that our illustration provides two (2) ways of presenting the IPOs. Students are confused because some thesis panels require the first while other requires the second. Both are correct.

Before writing the theoretical and conceptual framework, the researcher must first read and study some literatures and previous studies related to his study. From these literature and studies, the researcher may now formulate a theoretical scheme for his research problem. This scheme is a tentative explanation or theoretical explanation of the phenomenon or problem and serves as the basis for the formulation of research hypotheses. Thus, the theoretical/conceptual framework consists of the investigator's own position on a problem after his exposure to various theories that have bearing on the problem.

Example 3. *Conceptual Framework of the game: “The Mythical Adventures of Mang Kanor: An Action-Adventure Game”, are presented in this wise:*



Evaluation

To create successful 3D computer game, knowledge, skills, hardware, and software are needed. Since The Mythical Adventures of Mang Kanor is focused on mythical creatures and need to be built in a computer/laptop with Dual core processor or higher, familiarity in the mythical creatures and computer game development is needed.

The development of the project requires idea about the mythical creatures, Javascript, Kinematic Movement Algorithm: Arriving behavior, Steering Behavior, Wandering behavior and Path Following Algorithm. To come up with a working game, it is also required to have a hardware and software.

The required hardware is personal computer or laptop with physical memory at least 1GB for creating, testing and building the game. The software needed the Unity 3.5.5 with IDE mono develop that used for creating terrain, import characters from 3DS Max, scripting and building the whole project and Adobe Photoshop CS3 was used for editing textures for mapping characters, other 3D objects and for GUI interface design.”

2.6 Definition of Terms

This states the conceptual and operational definition of terms. Citing of source(s) should be in APA format.

Guidelines in Defining Terms:

1. Only terms, words, phrases, which have a special or unique meaning in the study, are defined.
2. Terms should be defined operationally, that is, how they are going to be used in the study.
3. The researcher may develop his/her own definition from the characteristics of the term defined. Thus, the level of efficiency of a system may be defined as a researcher-made scale to measure the efficiency.
4. Definitions may be taken from encyclopedias, books, magazines, newspaper articles, dictionaries, and other publications but the researcher must acknowledge his/her sources. Definitions taken from published materials are called conceptual or theoretical definitions.
5. Definitions should be brief, clear, and unequivocal (no double meanings).
6. Acronyms should always be spelled out fully especially if it is not commonly known or if it is used for the first time.

There are two ways of defining the key terms of the study:

Technical Definition – the meaning of the terms is taken from the dictionary or encyclopedia.

Operational Definition – the meaning of terms is based on an observed characteristics and how it is used in the study.

Example: (Source: Porras, et.al. E-Faculty Portfolio, SICT, WVSU Calinog Campus, Unpublished undergraduate research, 2015)

Faculty. *This refers to the entire teaching staff of a university, college, or school, including any administrators holding academic rank (Microsoft Encarta, 2008).*

In this study, the term “Faculty” refers to the entire teaching staff of the School of Information and Communications Technology (SICT) of the West Visayas State University – Calinog Campus.

Portfolio. *This refers to a collection of work (as papers and tests) compiled over a period and used for assessing performance or progress (Merriam-Webster Dictionary, 2012).*

In this study, the term “Portfolio” refers to a collection of documents pertaining to the SICT faculty. The SICT faculty portfolio contains documents such as: certificates, CHED-E0, class schedule, employment contract, diploma, designation, transcript, personal data sheet, service record, and syllabus. Moreover, the portfolio contents are bases of reflecting the competence of the faculty members which they may use for promotion or during the accreditation period.

2.7 Significance of the Study

States the significance or benefits of your system, both quantitative and qualitative. Specifically, what benefits your system will provide to the company, organization, department users, clients, your group and future researchers.

Tips in Writing the Significance of the Study

1. Refer to the Problem Statement

In writing the significance of the study, always refer to the statement of the problem. This way, you can clearly define the contribution of your study. To simplify, your research should answer this question, “What are the benefits or advantages of the study based on the statement of the problem?”

If you ask the question “How has the new packaging affected the sales of the product?” then the contribution of your research would probably a packaging style or technology that can help the store increase its sales. Your study should demonstrate that the product’s packaging really influences the buyer’s perception and affects their purchase decision.

2. Write it from General to Particular

Determine the specific contribution of your thesis study to society as well as to the individual. Write it deductively, starting from general to specific. Start your significance of the study broadly then narrowing it out to a specific group or person. This is done by looking into the general contribution of your study, such as its

importance to society as a whole, then move towards its contribution to individuals as yourself as a researcher.

Example 1: (Source: Gecole, G. et. al., Hardwired PAAB System of West Visayas State University- Calinog Campus, SICT, WVSU-CC, unpublished research 2014)

Significance of the Study

This proposed system would be of great help to the whole campus particularly to the faculty and staff as well as students of West Visayas State University Calinog Campus. The realization of this new system would help the old procedure to become easy, accurate, faster, reliable and especially it would be an effective system. This would be of great help to store important information during paging announcement, alarm and bell system.

Students. They would be benefitted of this capstone project for they would be provided an automated and effective school bell, paging, announcement and alarm for them to be attentive and alert.

Faculty and Staff. This proposed system would aid them to be more aware and conscious of the time frame allotted to the subjects and works. The Paging system would ease the spreading of useful messages and announcements among them.

Campus administrators – This would cut the cost of hiring a person to manually operate the traditional alarm system, provide flexibility during changes in time schedule as needed and ease his job of calling personnel, students, staffs, teachers etc. if the need arises.

Future researcher - This would also facilitate the future researchers because this would serve as reference related to their studied system.

Example 2: (Source: Orbino,B., et.al. Agri-Hayop-Alalay (AHA) Mobile Application, SICT, WVSU-CC, Unpublished Research, 2019)

Significance of the Study

Agri-Hayop-Alalay (AHA) Mobile Application aims to give an on-hand mini-animal doctor for livestock health consultations. Due to the fast growing of technology, anyone carrying a smart phone can almost do everything in the world with a little bit of tapping and then accurate results are obtained. This app helps the farmers about the most common health issues that a typical livestock animal can have. The application can be useful to the context below.

Farmer. The farmer is the main client of the proposed system. This will help them to have a thorough diagnosis or check-up of the problems of their livestock. This will also provide them useful information about livestock farming and will give them a hassle-free work of finding available technicians to check for their livestock animals.

Department of Agriculture. The Department of Agriculture can benefit from this kind of mobile application. If the farmers have this kind of application and the results can solve their problems they tend to not ask for a request for check-up, and it gives this department a minimal work load.

Community. The mobile application can be used by anyone; the community can get information about common animals that can be bred in this town. It is also beneficial for all that need clarifications about animals.

Researchers. This mobile application can be used by any researcher who wants to have information about livestock animals here in Calinog. This mobile app can also help them know more about livestock farming.

Students. The mobile application is informative about the common animals in Calinog. It also has the function of diagnosing some symptoms with recommendations or results that can help the farmers or any users that is using the application.

Future Researchers. All information and facts that are in the application may be of use by the future researchers for any purpose that may help them.

2.8 Purpose and Description

(Laviña, 2016), in writing the purpose and description, the proponent must state the function of his capstone project. It should not attempt to claim that the project will add to the body of knowledge. It is for this reason that the authors humbly submit, that members of the panel of examiners should not ask the proponent(s) the following question:

“What is the contribution of your capstone project to the body of knowledge?”

This is because IT and IS capstone projects generally, do not contribute to the body of knowledge but focuses on the contribution of the project to its specific beneficiaries.

Hence, the importance of the capstone project must contain explanations or discussions of any or all of the following:

1. The rationale, timeliness and relevance of the study.
2. Possible solutions to existing problems or improvement to unsatisfactory conditions.
3. The beneficiaries and how they are going to be benefitted.
4. Possible contribution to the fund of knowledge (if any).
5. Possible implications.

It is good to start the purpose and description by providing the function of the capstone project.

Example:

In the capstone project entitled, “Development and Evaluation of Classroom Schedule Generator (CSG): An Expert System Using Microsoft Visual Basic” the Purpose and Description was written as follows:

Purpose and Description

The project was conducted in order to assist the school administrators in the preparation of classroom schedules as well as teaching loads of faculty members every term or semester by developing a software called Classroom Schedule

Generator (CSG). Once developed, CSG shall automatically prepare the classroom schedules of students and teaching loads of faculty members.”

After providing its function, the second paragraph should provide description of the project by enumerating its features and capabilities. The following paragraph is a good illustration, thus:

CDS, an expert system have the following capabilities:

- 1. Assign classes, lecture or laboratory, in appropriate rooms. Hence, lecture classes shall not be assigned to laboratory classrooms and laboratory classes shall not be assigned to lecture classes.*
- 2. Maximizes room, student and faculty schedules depending on the priority set by the users. Hence, if there is a shortage of faculty member, say, only one teacher can handle a particular course, then the highest priority shall be set to faculty. On the other hand, if there is only one computer laboratory room, the highest priority shall be set to the said laboratory room.*
- 3. Assigned teaching load of faculty members in accordance to the course and time schedule preference set by the users. So, in case of part time faculty members, is only available T-TH 5:00-9:00PM, the system cannot assign classes to him beyond the said scheduler;*
- 4. Ensure that only four(4) preparations shall be given to any faculty members;*
- 5. Provide full load for fulltime faculty members depending on the number of units set by the user. So, if fulltime faculty members is teaching 18 units, 18 units shall be set; if the teaching units is set beyond 18 units, it means that the dean is allowing the said faculty members to have overload.*
- 6. Make an appropriate recommendation in case there will be not enough resources in making schedules.*

In writing the “**Purpose and Description**”, the features and capabilities should be written in such a way that it would highlight the significance of the project. Furthermore, it should be written using a general language and non-technical terms. This is to distinguish it from “**Requirement Documentation**” which will be discussed in Unit 4.

After enumerating the project’s features and capabilities, the last paragraph should explain project’s rationale, timeliness and relevance by illustrating that it will solve existing problems. Possible beneficiaries as well as implications may also be given. Next example, this will provides a good example of how the specific beneficiaries will be benefitted by the project.

Example 2. “Acceptability of the Multipurpose Community Information and Telecenter (MCIT) prototype towards E-Governance and Bridging the Digital Divide”

Purpose and Description

Information and Communications Technology (ICT) usage particularly the Internet has become a valuable resource throughout the country. Its use has forever changed the way people communicate, do business and educated. Physical distances between people have been narrowed as the Internet has made it possible to reach people almost instantly. At the same time, grave inequities exist in terms of who actually has access to Internet services. This discrepancy in access is called the Digital Divide. Data shows that the Internet and computer technologies are available far more to wealthy and educated people than they are to the poor and to racial minorities. The purpose of the project to bridge the so called digital divide.

- 1. National Economy. The Digital Divide is an essential issue to be addressed by the study because not all citizens are able to enjoy the benefits that come from ICT usage particularly the Internet. The Internet is an important resource that has grown because of its universality, yet there are millions of people who simply cannot access the new technology because of its cost. The Internet, then, is an essential good because of the role it plays in communication and commerce worldwide. However, if the internet remains an unattainable good to citizens of lower socioeconomic status, it could intensify current social divisions, resulting in wider economic inequities. Establishing Community Telecenters to every barangay which typically provide public access to computers, the Internet, and other communication technologies, is the first step towards bridging the digital divide.*
- 2. Commission on Election (COMELEC). During every election, registration and validation of every citizen of the country shall no longer be needed. The Commission on Elections (COMELEC), after an appropriate process, can retrieve the records of every citizen for each barangay. In addition, double registration is not possible in an MCIT enabled barangays. This is because the MCIT prototype uses biometrics technology particularly the fingerprint of an individual as key for identification. Henceforth, the integrated system can easily detect multiple fingerprints in different barangays .*
- 3. National Statistics Office (NSO). The regular census conducted by the National Statistics Office for each barangay in the whole country shall be minimized. This is because MCIT barangays have updated records therefore, the NSO can retrieve the database instead of doing a costly census usually conducted by inexperienced workers. The amount of money that shall be saved by the NSO and COMELEC when MCIT system is implemented for each barangay in the country cannot be discounted.*
- 4. Local Government' Units (LGUs). With the proposed MCIT system, Chapter 5, Section 394 (d) (6) of the local government code which states "The barangay secretary shall keep an updated records of all inhabitants of the barangay containing the following items of information: name, address, place and date of birth, sex, civil status, citizenship, occupation, and other items of information as may be prescribed by law or ordinance" shall be fully implemented. Aside from being complaint with the law, MCIT enabled barangays will be effective in managing information. The computer based information system will provide information to the concerned individual/ barangay officials in the proper form at the proper time. This will ensure that legal documents acquired from barangay are reliable, accurate and dependable. Furthermore, the study will also provide the concerned LGU with basic knowledge management skills since knowledge mapping can be done using MCIT system. The system can be used to retrieve the list of experts from a certain LGU. This will help*

the LGU from barangay level to the provincial level in seeking professional assistance for their project. The system in return will help the knowledge workers in earning more income.

5. *The General Public. Since MCIT barangay is not an information center but also a Telecenter, the study will make MCIT- enabled barangay as an outlet for the general public with affordable access to a variety of services using ICTs especially from government Online services such as NSO (e-Census), SSS (on-line inquiry), GSIS (e GSIS), DTI (i-Reklamo Online Consumer Assistance) DBM (Electronic Tin Verification) and many Others.*
6. *Researcher. Since the study looked into the aspects of information management resources and practices, extent of ICT usage, the level of the acceptability of the MCIT prototype and the factors that may affect the respondent's level of acceptance involving a total of one hundred seventeen (117) respondents from different barangays from selected municipalities of the Province of Laguna, the Research, IT Management and Decision Making capabilities of the researcher will surely be improved. In addition, the researcher was able to personally discern the problems met by concerned barangays and apply the right technology and management principles that would resolve the present unsatisfactory conditions. Furthermost, the researcher became one of the knowledgeable persons in the establishment of Multipurpose Community Information and Telecenter which in return will prepare him as future Information Technology Consultant and/or Manager.*
7. *Future Researchers. Before the establishment of MCIT in the different barangays in the country, its level of acceptability must first be known. The cost of establishing community telecenter must be known vis a vis the financial capabilities of the LGUs concerned. The findings and conclusions of the study will be significant to both allies and critics of Community Telecenter Movement.*

Obviously, the proposed study is a response to the challenge of bringing the rural villages out of isolation by making the information society to be more about inclusion rather than exclusion. Henceforth, the study will help in alleviating the digital divide in the country if not in the whole world.

2.9 Scope and Limitations of the Study

The section may be subdivided into two parts, the Scope and Delimitation and Limitation of the Study.

The Scope and Delimitation of the Study should include a brief statement of the general purpose of the study, the target users/beneficiaries of the study, the period of the study and the features of the proposed software.

The limitation of the study includes the weaknesses of the study beyond the control of the researcher. This section may be not be included in the study if there is no such weaknesses of the study beyond the control of the researcher.

Example 1: (Source: Orbino,B., et.al. Agri-Hayop-Alalay (AHA) Mobile Application, SICT, WVSU-CC, Unpublished Research, 2019)

Scope and Limitations of the Study

The Agri-Hayop-Alalay (AHA) mobile application is educational or informative for farmers and others that are interested in livestock farming. The main function of the system is defining the symptoms of animals or diagnosing. The application has a number of animals that a user can choose or select for the diagnosis. The result or after describing what kind of symptoms of the animal will show, and the following information will be shown: (1) the disease of the animal and the information about it; (2) the suggested instruction or things to do to cure the disease. The proposed system also provides information about selected animals. The application is made for the farmer's assistance. Through the chatroom, farmers can communicate to any registered technician in the Municipal Agriculture Office of Calinog and registered farmers that are using this system.

The AHA mobile application focuses only on common livestock animals that farmers in the town of Calinog are raising, specifically carabaos, pigs, goats, chickens, and cows. Complex or rare cases of diseases are beyond the systems operations or capabilities. The application can be used by anyone capable of using a Smartphone. The chatroom in the proposed system can be accessed by anyone. Users cannot make a private message for every registered person that is using the system. The application is dependent type of system; it does need an internet connection to fully run or function.

Name:	Date:
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● **Review Questions:**

1. Explain the significant and relevant distinction of statement of the problem and objectives of the project.

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- ## 2. Differentiate Purpose and Description against Significance of the Study.

[illegible]

1. According to your approved title, write down the following:

a. Introduction

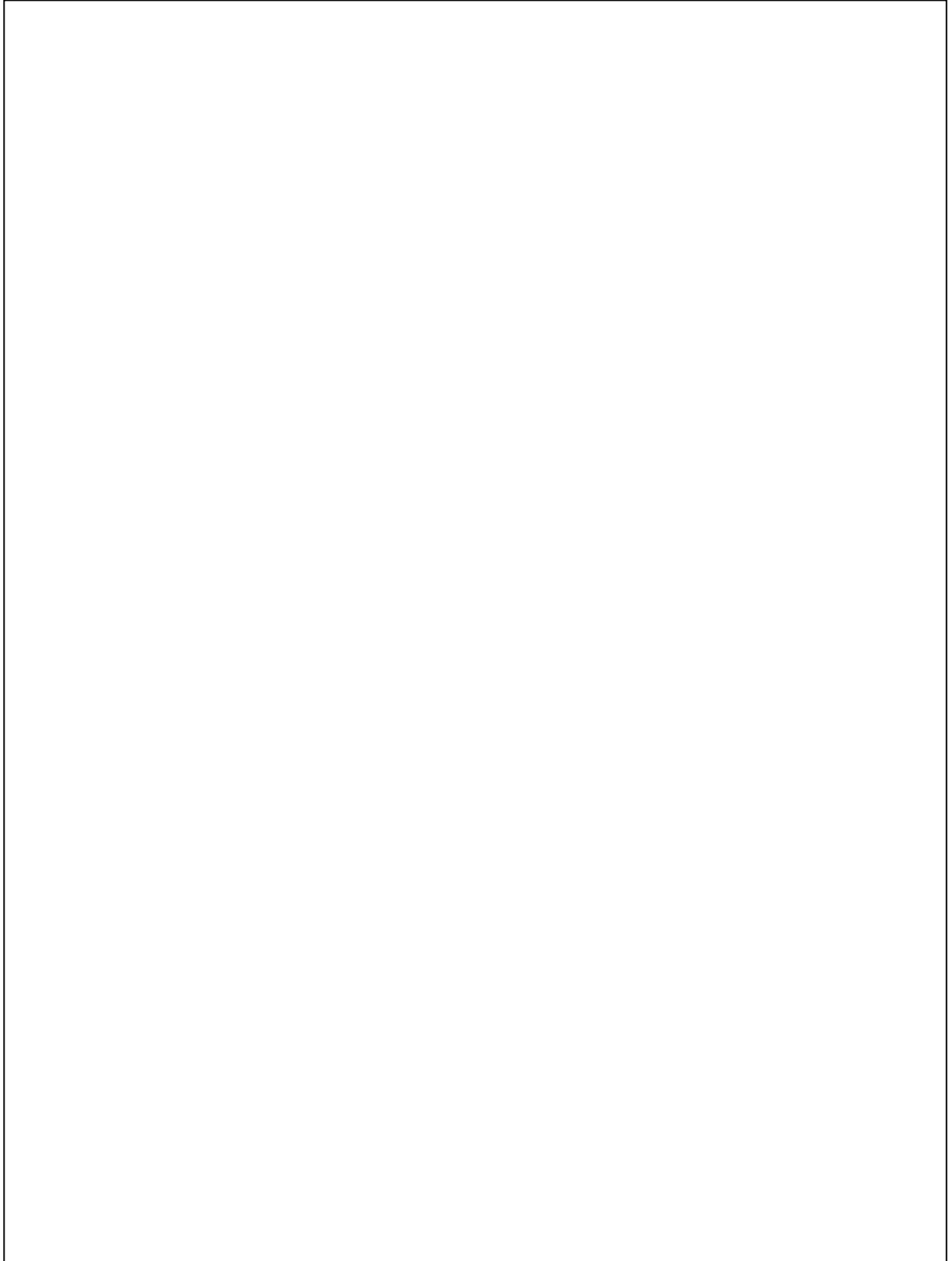
This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins or other markings on the paper.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on its right side, suggesting it's resting on a surface.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

e. Conceptual Framework



f. Definition of terms

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

g. Significance of the Study

[illegible]

[illegible][illegible]

● **References:**

Agham, R. et.al. Phil Pres Trevia, College Of Technology, ISCOF-Dumangas Campus, Unpublished Undergraduate Research, 2015

Calmorin L.P., Calmorin M.A. Methods of Research and Thesis Writing: Rex Book Store 2001

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Laviña, C. G. , et. Al., Outcomes-Based Practical Guide to Thesis and Capstone Project Writing in Computing, Mindshapers Co., Inc. 2016

Orbino,B., et.al. Agri-Hayop-Alalay (AHA) Mobile Application, SICT, WVSU-CC, Unpublished Research, 2019

Porras, et.al. E-Faculty Portfolio, SICT, WVSU Calinog Campus, Unpublished undergraduate research , 2015

UNIT-III: WRITING THE REVIEW OF RELATED LITERATURE AND SYSTEMS

INTENDED LEARNING OUTCOMES (ILOs)

At the end of this unit; the students must have:

1. described the technical background;
2. identified the objectives of citing Literature and System;
3. recognized the guidelines of citing literature and systems;
4. determined the relatedness of the present study to selected literature or system;
5. determined the instances when to quote or paraphrase sources;
6. avoided plagiarisms during the research process; and
7. developed and compiled an appropriate related literatures and studies.

3.1 Technical Background

A researcher needs to review write-ups, readings, and studies related to his present study. Funk and Wagnalls (1972) defines literature as “*Written works collectively, especially, those of enduring importance, exhibiting creative imagination and artistic skill, which are written in a particular period, language, and subject.*” In other words, any written materials such as book, journal, magazine, novel, poetry, yearbook, encyclopedia, and other similar written materials are considered literature.

Technical background includes an in-depth discussion on the relevant technical aspects of the project, the authors submit, that the theoretical/technical background should be placed before the review of related literatures proper. This is because the review of related literatures and studies contains what has been said and what has been done about particular subject of research. Hence, it is better to discuss first the technical or theoretical aspects of computing in order for the reader to gain substantial understanding about the subject matter before providing what has been said and what has been done about the said subject (Laviña, 2016).

Example: In the research work entitled “iBantay: An Interactive Cross Platform Mobile Application for Open Government Data Access”. Part of its Technical Background, discussing about Cross Platform Mobile Development reads:

Source: (Abarrintos, E. et.al., NCITE 2014)

Technical Background

Cross Platform Mobile Development

A recent survey among developers by Telerik showed that more developers are moving away from purely native development to multiple platforms because of the attractiveness and cost-effectiveness of Web technologies such as HTML5, CSS 3 and Javascript, Mobile applications developed using these technologies can rival native apps in terms of rich user experience and access to advanced capabilities like storage and geo-location. Cross-platform mobile development also became possible through the

use of tools and frameworks such as JQM (Jquery Mobile) and PhoneGap that enable developers to implement mobile apps that can run seamlessly without modification on different platforms. The term cross-platform or interoperability can be classified into three: (1) those that exist within operating systems; (2) across operating systems; and (3) across devices. Cross platform development uses a technique called “cross compilation” that separates the build environment from the target environment. In mobile development context it works as follows: (1) the framework (e.g. PhoneGap) provides platform independent API (application programming interface) using mainstream programming language (e.g. JavaScript, Ruby, or Java); (2) developers use the API to build the mobile application, including the UI (user interface), data persistence and business logic; (3) the code is then processed by a cross compiler that transforms it into platform-specific native app(s) targeted at the different platforms that the application will run on.

Another increasingly popular approach in cross-platform development is to build a mobile app that will run on the user’s mobile browser. This involves using standard web technologies like HTML, CSS and Javascript to build application and make it look and behave like a native app. This is possible due to the advanced capabilities of HTML5 and CSS 3, including embedded SQL databases, local storage, animations, canvas, web sockets and video playback.

Cross platform development uses a variety of frameworks and tools that can be divided into high-level groupings:

- *Library. Small, self-contained toolkit that offers very specific functionality to developer (e.g. UI widgets and 3D graphics libraries).*
- *Framework. A set of libraries, software components and architecture that provides developer with a comprehensive toolkit to build a complete mobile app.*
- *Platform. A set of frameworks, tools and services that not only allow developers to build a complete mobile app but also to configure, package and distribute it to app stores or the cloud.*
- *Product/Service. Offers a specific functionality or service ready to be used and integrated into a mobile solution.*

3.2 Review of Foreign or Local Literature

Related Literature is comprised of discussions of facts and principles to which the present study is related. For example, if the present study deals with web application development, literature to be reviewed or surveyed should comprised of materials that deal with web application development. These materials are usually printed and found in books, encyclopedias, professional journals, magazines, newspapers, and other publications.

These materials are categorized as

1. Local, if printed in the Philippines; and
2. Foreign, if printed in other lands.

3.2.1 Aim of Literature Review

The aim of a literature review is to show the reader what the researchers have read, and have a good grasp of, the main published work concerning a particular topic or question in the specific field.

It is very important to note that the review should not be simply a description of what others have published in the form of a set of summaries, but should take the form of a critical discussion, showing insight and an awareness of differing arguments,

theories and approaches. It should be a synthesis and analysis of the relevant published work, linked at all times to your own purpose and rationale.

3.2.2 Elements of Literature Review

Literature reviews should comprise the following elements:

1. An overview of the subject, issue or theory under consideration, along with the objectives of the literature review;
2. Division of works under review into categories (e.g. those in support of a particular position, those against, and those offering alternative theses entirely);
3. Explanation of how each work is similar to and how it varies from the others;
4. Conclusions as to which pieces are best considered in their argument, are most convincing of their opinions, and make the greatest contribution to the understanding and development of their area of research.

According to Caulley (1992) of La Trobe University, the literature review should:

- compare and contrast different authors' view on an issue
- group authors who draw similar conclusions
- criticize aspects to methodology
- note areas in which authors are in disagreement
- highlight exemplary studies
- highlight gaps in research
- show how your study relates to previous studies
- show how your study relates to the literature in general
- conclude by summarizing what the literature says

In a broader context Hart(1998) lists the following purposes of a review

- Distinguishing what has been done from what needs to be done;
- Discovering important variables relevant to the topic;
- Synthesizing and gaining a new perspective;
- Identifying relationships between ideas and practices;
- Establishing the context of the topic or problem;
- Rationalizing the significance of the problem;
- Enhancing and acquiring the subject vocabulary;
- Understanding the structure of the subject;
- Relating ideas and theory to applications;
- Identifying methodologies and techniques that have been used;
- Placing the research in a historical context to show familiarity with state-of-the-art developments.

Example: (Source: Agham, R. et.al. Phil Pres Trevia, College Of Technology, ISCOF-Dumangas Campus, Unpublished Undergraduate Research, 2015)

Review of Related Foreign and Local Literatures

Trivia 360

Trivia 360 is a free quiz game you can play on your android. It is an easy to play trivia app but at the same time it is an excellent brain game. It also gives the brain an immaculate boost by addictive thinking and IO challenge. The quizzes are

set in different categories such as Classic 4-answer questions, true/false questions, flag quiz, landmark riddles and more (<https://www.appgroves.com>, 2019).

In relation to the study, Phil Pres Trivia is also applicable and can be played on android phones. Both games are having similar in some ways on how to boost the thinking and knowledge of a user. Trivia 360 game has a topic about the Presidents but its focus does not only contain in the Philippines rather in the whole continent whereas, Phil Pres Trivia has its focus in the Presidents of the Philippines only.

Philippine Presidents Quiz

Philippine Presidents Quiz is an app that test if how well you know the Presidents of the Philippines. This is a good game that can be played with kids to teach them about the country and its history and to learn about the names of the Presidents. It displays four images of Philippine Presidents as the choices and correct answer will display the name of the President. It also has the Presidents information and focuses only on familiarizing the images likewise, the Presidents names (<https://play.google.com>, 2017).

In relation to the study, both games are related when it comes to its content. Both of it has the same topic to the Presidents of the Philippines but the different is that Philippine Presidents Quiz focuses only on familiarizing the Presidents name and images whereas, the Phil Pres Trivia focuses on Presidents slogan, quotes, images and biography.

Philippines Quiz

A wonderful trivia game for the pinoy people and Filipino culture. This game will test the knowledge of a user through given a hint and the user will manually encode the best answer to the question. It contains questions about sport, history, geography, general knowledge about the Philippines. The user will have fun while learning more about the Philippines (<https://play.google.com>, 2016).

In relation to the study, Philippine Quiz contains random questions about the Philippines that includes sport, history, geography, and general knowledge. On the other hand, Phil Pres Trivia focuses only on the information about Presidents of the Philippines.

3.3 Review of Foreign or Local Systems

A published as well as unpublished research study that focuses on the systems similar to the study being conducted is sources of materials which may be included in this section. The research studies may also be identified as foreign or local.

Example: (Source: Agham, R. et.al. Phil Pres Trevia, College Of Technology, ISCOF-Dumangas Campus, Unpublished Undergraduate Research, 2015)

Review of Related Foreign and Local System

4 Pics 1 Word

4 Pics 1 word is a game with a very simple pre. It displays 4 photos which players need to use to guess if which word is hidden among the different available letters. The puzzle will usually is hidden among the different available letters. The

puzzle will usually be extremely difficult. So the players have to use two different shortcuts. By completing the puzzle correctly, the players will earn coins. The best thing about 4 Pics 1 Word is the fact that it is a very simple game with very basic mechanics. Anytime the player can enjoy the game whether it is for a few minutes or a number of hours (<https://4-pics-1-word.en.uptodown.com>, 2013).

In relation to the study, 4 Pics 1 Word is a simple, fun game that will enjoy of either having on your own or with friends. While the “Phil Pres Trivia”, anyone can have it as long as they have their own android phones. It is composed of a Presidents’ slogan, quotes, guess who and Presidents’ biography that tends to give information and also to test the knowledge of a user.

QUIZOID: Trivia Quiz 2016

Quizoid: Trivia Quiz 2016 is a game that tests the knowledge of a player across a variety of subjects and genres. This trivia quiz game offers over 6, 000 multiple – choice questions across 17 different fields, including General Knowledge, Science, Math, Politics and Religion. It can be played offline, and when online, questions are updated regularly for the players to have a new bit of trivia to train their brain (<https://downloads.tomsguide.com>, 2015)

In relation to the study, “Phil Pres Trivia” is divided into four category, the Presidents slogan, Presidents quotes, Guess who that lets the user to guess the Presidents name through the given image and also, it has Presidents biography that store the information of Philippine Presidents. It can also be played offline.

Quiz of Knowledge

Quiz of Knowledge is the ultimate trivia quiz because it contains more than 4.000 questions. There are questions about Geography, Sports, mythology, celebrities and so on. The questions belong to 16 categories and are updated on a weekly basis. The categories are as follows: History, Geography, Theater, Music, Science, Religion, Arts, Space, Sports, World Mythology, Science, Art, Technology, Animals, Politics and Language. Unlike many other trivia games Quiz of Knowledge is playable offline anytime (<https://play.google.com>, 2019).

In relation to the study, Phil Pres Trivia can also be played offline and the game has also category. The difference is Quiz of Knowledge contains more than 4, 000 questions categorized 16 subjects whereas, Phil Pres Trivia has four category that deals only in Philippine Presidents.

3.4 Different Ways of Citing Literatures and Systems

Citing Sources In-Text

2. Whenever you quote, paraphrase, summarize or otherwise refer to the work of another, you are required to cite its source by the way of parenthetical documentation.
3. The authors recommend the American Psychological Association (APA) guidelines and the Chicago Manual of Style. These guidelines require that you use parenthetical citations to document quotations, paraphrases, summaries, and other material from a source used in your paper. These in-text citations correspond to the full bibliographic entries found in a list of references included at the end of the

paper. Unless otherwise noted, electronic sources follow the same pattern as printed ones.

4. To cite specific part of a source, indicate the page, chapter, figure, table or equation, at the appropriate point in the text. Always give page numbers for quotations. Note that the words page and chapter are abbreviated in such text citations:

(Cheek and Buss, 1981, p.332)

(Shimamura, 1989, chap.3)

Single author named in a signal phrase.

Use the author's name in a signal phrase to introduce the quoted material, and place the date of the work in parentheses, immediately after the author's name.

For a quotation, the page number, preceded by p., appears in parentheses after the quotation.

Example:

Social historian Richard Sennett (1980) names the tendency to come to terms with difficult experiences a "purification process" whereby "threatening or painful dissonances are warded off to preserve intact a clear and articulated image of oneself and one's place in the world" (p.11).

Single author named in parentheses.

When you do not mention the author in the signal phrase, give the name and the date, separated by the comma, in parentheses at the end of the cited material.

Example:

The tendency to come to terms with difficult experiences a "purification process" whereby "threatening or painful dissonances are warded off to preserve intact a clear and articulated image of oneself and one's place in the world" (Sennett, 1980, p.11).

Two Authors

Example:

Goody and Watt (1963) have gone so far as to declare that "the most significant elements of human culture are undoubtedly channeled through words, and reside in the particular range of meanings and attitudes which members of any society attach to the verbal symbols" (p. 323).

Three to Five Authors

List all the authors names for the first sentence.

Example:

Kintgen, Kroll, and Rose (1988) maintain that "just as a single definition of literacy is insufficient, so is scrutiny from within the confines of a single academic discipline" (p.xv)

In subsequent references, use just the first author's name plus et al.

Example:

In assessing the educational quality of our schools, it is important to remember that, as Kintgen et al. (1988) explain: “the contemporary asymmetry between reading and writing can be related to use in a particular socioeconomic context” (p.xvii).

Six or more authors

Example:

As Williams et al. (1999) demonstrated, the internet holds the potential to open consumer markets in new and unexpected ways.

Corporate author (organization, association, etc.)

If the name of the organization or association is long, spell it out the first time, followed by an abbreviation in brackets. In later citations, use the abbreviation in brackets. In later citations, use the abbreviation only.

Example:

FIRST CITATION (Food and Drug Administration [FDA], 1996)
LATER CITATION (FDA, 1996)

Works with no author

Example:

Several critics of the concept of the transparent society ask if a large society would be able to handle the complete loss of privacy (“Surveillance Society,” 1998, p. 115).

Bible passage

Example:

Unfortunately, the president could not recall the truism found in Prov. 20-22 that “Wisdom is a fountain to one who has it, but folly is the punishment of fools” (New Oxford Annotated Bible).

E-mail message

Example:

Robinson’s argument has since been dismissed by scholars of American history (Virginia Baker, personal communication, October 28, 2002).

Web page

For electronic sources that do not provide page numbers, use the paragraph number, if available, preceded by the paragraph symbol or the abbreviation para. If neither paragraph nor page numbers are visible, cite the heading and the number of the paragraph following it to direct the reader to the location of the material.

Example:

(Myers, 2000, #15)

(Beutler, 2000, Conclusion section, para. 1)

The cabin at the Abraham Lincoln Birthplace National Historic Site is designed to be “symbolic of the one in which Lincoln was born” (National Park Service, 2003, para. 1)

Note: page or chapter number should be included if available on the web site.

3.5 Justification of the Present Project (Synthesis)

In this section, the researcher should be able to convince his reader that his thesis or capstone project is not a duplication of other's work. In case that the proposed thesis or capstone project is a continuation of a previous work, this section should give emphasis or justification why the proposed thesis or capstone project as the case, maybe is needed. Example of a good justification is when the previous work recommends that further improvement is needed to maximize the benefits of his work.

There may also be a need to continue with the present investigation to affirm or negate the findings of other inquiries about the same research problem or topic so that generalizations or principles may be formulated. These generalizations and principles would be the contributions of the present investigation together with other studies to the fund of knowledge.

Example:

In the research work entitled, “STUDDDEDNOTES: An Automatic Transcription of Melody and Chords from Singing Voice into Lead Sheet Using Melody Extraction and Chords Detection”, by: Dadiz, B. et al., NCITE, 2014)

Synthesis

Both papers (here they are referring to those two-research mentioned in the Related Studies) have similar approach in generating lead sheets. They used chord detection algorithms and melody extraction algorithms to determine the structure of the audio. Moreover, their study also used beat modules to align the chord correctly. The system presented by J Weil et. al., 2008 is less accurate than the one proposed by T. Sikora et.al., 2009 as the former lack silence detection.

Compared to the two algorithms given by the two research studies on extracting melody, the algorithm proposed by J. Salamon et. al., 2012, is more accurate. This paper's algorithm has also better accuracy than the current state-of-the-art algorithms. In chord detection, the Hidden Markov Model performs poorly than the hypothesis-testing algorithm. YIN algorithm is used because of its simplicity and usefulness for many applications. The algorithm of J. Salamon, et.al., 2012 is used for polyphony, which is not the scope of this research. On the other hand, the Hidden Markov Models needs a lot of training data as dataset for it to distinguish human voice that is why this study have not implemented HMM. This study already determines the notes, using the notes result for harmonizing the chord is makes the result more accurate.

In both systems for generating music lead sheet, all of them have their audio that is recorded in a studio like environment. Recordings made, for example in ordinary surroundings where not included in the test data so the accuracy on that part is not known. The study of J-L. Durrieu, et.al. the melody and chords detection

module does not talk with one another. Their connection would lead better result. The melody module does not detect the silence part of the recording. This leads to a lead sheet that contains only pitch notes without any rest note. This is not sufficient when being used by musicians. Another paper by T. Sikora, et.al. (2009) propose a system that would also generate lead sheet. They use various algorithms from the literature make extraction of harmonic features for the lead sheet. The paper uses songs from the Western pop culture and synthesized audios are used. It is not known how well it would perform when the audio came from the ordinary recording of the user who has no luxury of having a music studio.

Because of these findings, the study of generating lead sheet is proposed to fill up these deficiencies. It would take the recording of the user using the device available to him and would generate lead sheet afterwards. The melody extraction will detect the part of the recording that has silence. Rest notes would be encoded to the generated lead sheet making it sustainable for musicians use. The pitch detection in the melody extraction module is known to be a good in vocal and non-vocal sound increasing the confidence of in the reliability of the melody extracted. The interconnection of the melody and the chords module in the proposed system made the chords detected harmonize with the detected melody. Usually, the melody is the basis for chord harmonization.”

3.6 Common Guidelines in Citing Related Literature and Studies

The following are the characteristics of related literature and studies that should be cited:

- The materials must be as recent as possible.
- Materials must be objective and unbiased as possible.
- The materials must be relevant to the study.
- Materials must not be too few but not too many.
- After the reading, the readers should gain adequate understanding on the technical topic(s) involved in the thesis and /or capstone project.

3.7 When to Quote, When to Paraphrase

The real “art” to research writing is using quotes and paraphrases from evidence effectively in order to support the researcher’s point. There are certain “rules”, dictated by the rules of style which must be followed, such as the one’s presented by the Modern Language Association (MLA) or the one’s presented by the American Psychological Association (APA). There are certain “guidelines” and suggestions.

But when all is said and done, the question of when to quote and when to paraphrase depends a great deal on the specific context of the writing and the effect the researcher is trying to achieve. Learning the best times to quote and paraphrase takes practice and experience.

In general, **it is best to use a quote when:**

1. **The exact words of the source are important for the point the researcher is trying to make.** This is especially true if you are quoting technical language, terms, or very specific word choices.

2. **The researcher wants to highlight his agreement with the author's words.** If the researchers agree with the point the author of the evidence makes and like their exact words, the researcher must use them as a quote.
3. **The author wants to highlight his disagreement with the authors words.** In other words, the researcher may sometimes want to use a direct quote to indicate exactly what it is he disagrees about. This might be particularly true when researchers are considering the anti ethical positions in his research writing projects.

In general, it is best to paraphrase when:

1. **There is no good reason to use a quote to refer to the evidence.** If the authors exact words are not especially important to the point the researcher is trying to make, it is usually better to paraphrase the evidence.
2. **Trying to explain a particular a piece of evidence in order to explain or interpret it in more detail.** This might be particularly true in writing projects like critiques.
3. **A need to balance a direct quote in writing.** Researchers need to be careful about directly quoting in their researches too much because it can sometimes lead to awkward and difficult to read pose. So, one of the reasons to use an paraphrase instead of a quote is to create balance within the researcher's writing.

3.8 Plagiarism

A common definition of plagiarism is the act of taking credit for someone else work. According to Supreme Court Justice Isagani Cruz, "*Plagiarism is passing off as your own somebody else's ideas or words.*"

In **plagiarism** the culprit claimed the work as his. If the culprit reproduced the copyrighted work, which he plagiarized, he is now guilty of both plagiarism and copyright infringement.

3.9 Copyright Infringements

This copyright Infringement is within the LEGAL aspect while plagiarism is within the area of ETHICAL aspect.

In copyright infringement, the culprit is NOT claiming the work as his, but he is able to make money out of it by doing acts, which only the copyright holder can legally do. A good example is the unauthorized reproduction of a copyrighted work and selling it for a lower price.

Penalty for copyright infringement is provided by law. The person who will acquire the infringed object/work shall also be liable. This is because the buyer is in bad faith because the infringer did not claim nor assert that he is the copyright holder or an authorized distributor/ seller. In other words the buyer is aware that he is buying a pirated item. For example, if Juan went to Downtown to buy a pirated movie "To Kill", Juan is aware that he is buying a pirated DVD. In criminal law, when a buyer bought a stolen property, say, jewelry for a lower price, the buyer is an accessory to the crime of theft. They are considered accessory to the crime because they also benefitted because they were aware that such was a stolen property and for that reason, they bought it at such a lower price. Under the anti-fencing law (PD 1612), the crime is not accessory but already a principal offense.

Furthermore, the REPUBLIC ACT NO. 9239 otherwise known as “The Optical Media Act” was enacted on February 10, 2004 to strengthen the fight against the proliferation of pirated CDs in the market.

In plagiarism, in case the culprit would sell the object plagiarized, the buyer may not be liable because of good faith. To illustrate, If Ana sold stolen jewelry to Pedra but Ana claimed that the jewelry was his and so Pedra was able to sell the jewelry at a reasonable price, Pedra will NOT be liable.

So, to avoid plagiarism, we should always attribute the work, quotations, etc., to its proper author. If we are not sure who the real author is, cite your reference. If forget the source but you can still remember the quotations, make a disclaimer or put anonymous. It is because when researcher does not attribute a phrase, paragraph to someone else, the presumption is, he is the author (Laviña,C., et. al. 2016).

Name:

Date:

● **Review Questions**

1. Why related literature important in research?

2. What is the nature of related literature and studies?

3. From what sources may related literature and related system be taken? Where are these sources located?

1. Give at least two (2) foreign and two (2) latest related system to your approved title, identify the areas of controversy or challenge and explain how your approved capstone project title will respond to the challenge and in the last paragraph how it is related to your study. Provide additional paper for your answer. It will be strictly handwritten.

This image shows a full page of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.

2. Give at least two (2) foreign and two (2) latest related literature to your approved title, identify the areas of controversy or challenge and explain how your approved capstone project title will respond to the challenge and in the last paragraph how it is related to your study. Provide additional paper for your answer. It will be strictly handwritten.

[illegible]

References

Agham, R. et.al. Phil Pres Trevia, College of Technology, ISCOF-Dumangas Campus, Unpublished Undergraduate Research, 2015

Laviña, C., et. Al., Outcomes-Based Practical Guide to Thesis and Capstone Project Writing in Computing, Mindshapers Co., Inc. 2016

Abarintos, E. et.al, iBantay: An Interactive Cross Platform Mobile Application for Open Government Data Access, NCITE 2014

UNIT-IV: Writing the Design and Methodology

INTENDED LEARNING OUTCOMES (ILOs)

At the end of this unit; the students must have:

- described the methodology;
- applied the appropriate methods in his/her capstone project undertakings;
- described in detail how the software will be designed;
- formulated the design of the system;
- created and write appropriately the Chapter III of the Capstone Project.

4.1 The Methodology

There are two main groups of research methods in the social sciences:

The empirical-analytical group approaches the study of social sciences in a similar manner that researchers study the natural sciences. This type of research focuses on objective knowledge, research questions that can be answered yes or no, and operational definitions of variables to be measured. The empirical-analytical group employs deductive reasoning that uses existing theory as a foundation for formulating hypotheses that need to be tested. This approach is focused on explanation.

The interpretative group of methods is focused on understanding phenomenon in a comprehensive, holistic way. Interpretive methods focus on analytically disclosing the meaning-making practices of human subjects [the why, how, or by what means people do what they do], while showing how those practices arrange so that it can be used to generate observable outcomes. Interpretive methods allow you to recognize your connection to the phenomena under investigation. However, the interpretative group requires careful examination of variables because it focuses more on subjective knowledge.

Content

The introduction to your methodology section should begin by restating the research problem and underlying assumptions underpinning your study. This is followed by situating the methods you will use to gather, analyze, and process information within the overall “tradition” of your field of study and within the particular research design you have chosen to study the problem. If the method you choose lies outside of the tradition of your field [i.e., your review of the literature demonstrates that it is not commonly used], provide a justification for how your choice of methods specifically addresses the research problem in ways that have not been utilized in prior studies.

The remainder of your methodology section should describe the following:

- Decisions made in selecting the data you have analysed or, in the case of qualitative research, the subjects and research setting you have examined,
- Tools and methods used to identify and collect information, and how you identified relevant variables,
- The ways in which you processed the data and the procedures you used to analyse that data, and

·The specific research tools or strategies that you utilized to study the underlying hypothesis and research questions.

In addition, an effectively written methodology section should:

Introduce the overall methodological approach for investigating your research problem. Is your study qualitative or quantitative or a combination of both (mixed method)? Are you going to take a special approach, such as action research, or a more neutral stance?

Definition

The methods section of the research describes the actions taken to investigate a problem and its rationale for the implementation of the techniques and procedures to determine the process and to analyze the information collected in order to understand the problem and to allow the readers to critically evaluate the study's validity and reliability. This section states the "how" the data was collected or generated and the "how" the data was being analyzed, at which, they are expressed in past tenses and must be direct and precise.

Importance of a Good Methodology Section

The researcher should explain how the data are obtained and analyzed the results to consider the readers must understand how the data are gathered since the method used affects the findings. Further, the readers should know how the gathered data were interpreted.

(<https://www.studocu.com/en-gb/document/university-of-essex/capstone-project/methodology-guidelines/1550026>)

The starting point in working with a capstone project is dealing with the review of the related literature in-order to know what has been already published and what something new can be developed. Scholars in the field have already used different methods at which are already valid and reliable, a point where your methods may be identified necessary in your capstone project development. (<https://capstone.extension.harvard.edu/>)

Researchers in the field of Information Technology considered the main point of ideas as the basis of generating solutions to the current problems and trends in technology era. Common method used in gathering the data is the conduct of interview to the company manager, operation and the end-user of the system where it will be utilized. It is also very important to know how needed the system is to be developed. In this case, "the need assessment" is highly recommended.

4.2 Requirements Analysis

A requirement is a condition and/or capability that conforms the system. Thus, the systematic approach in finding, documenting, organizing and tracking the series of change in the requirements of the system is known as requirements management.

The proposed system or software must provide computing solutions to address the needs of an organization. Thus, the need to determine information requirements about the specific organization under study will always be considered. At this phase, the details of the current system function: who (the people who are involved, what (the business activity), where (environment in which the work takes), when (the timing), and how (how the current

procedures are performed) of the business or organizations under study should be known by the researcher or the proponent of the study. Knowing and analyzing why the business uses the current system and a good reason why the organization is still using the current methods, should be considered in designing the proposed system. This includes the current technical situation.

After determining the information requirements and the proponent strongly believes that improvement may be done, he may propose the same to the organization under study. A data flow diagram and organization operational framework must be provided in this phase.

The kinds of requirements are categorized in two:

1. The User requirements. In this category, the needs and goals of the end-user are the priority. It includes what is expected to provide by the system, and the constraints on how it should work with its operations.
2. The System requirements. This category sends out the system's functions, services and operational constraints in detail.

Software system requirements are classified into functional and non-functional requirements:

1. Functional requirements. This specifies actions what a system must perform, without considering its physical constraints. Thus, it specifies the input and output behavior of a system.
2. Non-functional requirements. It describes the attributes of the system or attributes of the system environment. This is categorized into usability, reliability, performance, and substitutability requirements. Further, it specifies the need of compliance with any legal and regulatory requirements. It can also be a design constraint due to the operating system used, the platform environment, compatibility issues, or any application standards that apply.

Domain requirements is a subcategory of requirements. It comes from the application domain of the system and reflects characteristics and constraints of the said domain. It is formulated by domain-specific terminology and can be functional or non-functional requirements.

The functional requirements provide the user requirements. It is best described in a UML use-case model and in use cases. Every functionality of the system is modelled by a use case or even more.

The general objectives of the requirements discipline are:

1. To establish and maintain agreement with the customers and other stakeholders on what the system should do.
2. To provide system developers with a better understanding of the system requirements.
3. To define the boundaries of the system.
4. To provide a basis for planning the technical contents of iterations.
5. To provide a basis for estimating cost and time to develop the system.
6. To define a user-interface for the system, focusing on the needs and goals of the users.

In the frame of Rational Unified Process (RUP) methodology, the activities and artifacts are organized into workflow details as follows:

1. Analyze the Problem
2. Understand Stakeholder Needs

3. Define the System
4. Manage the Scope of the System
5. Refine the System Definition
6. Manage Changing Requirements

Each workflow detail represents a key skill that needs to be applied to perform effective requirements management. Analyze the Problem and Understand Stakeholder Needs focused mainly on the Inception phase of the project, whereas the emphasis is on Define the System and Refine the System Definition during the Elaboration phase. Manage the Scope of the System and Manage Changing Requirements are done continuously throughout the development project.

Analyze the problem

Problem analysis is done to understand problems, initial stakeholder needs, and propose high-level solutions. It is an act of reasoning and analysis to find "the problem behind the problem". During problem analysis, agreement is gained on the real problems and stakeholders are identified. Also, initial boundaries of the solution and constraints are defined from both technical and business perspectives.

The purpose of this workflow detail is to:

1. Gain agreement on the problem being solved,
2. Identify stakeholders,
3. Define the system boundaries, and
4. Identify constraints imposed on the system.

Glossary

The first step in any problem analysis is to make sure that all parties involved agree on what is the problem that we are trying to solve with our system. To avoid misunderstandings, everyone must agree on common terminology which will be used throughout the project in order to provide a consistent set of definitions for the system. Early on, we should begin defining our project terms in a glossary which will be maintained throughout the project lifecycle.

Actors

One of the most important considerations to identify in understanding the problem are the stakeholders who can best be represented by actors in a use-case model.

Use-case is referred to as the model of a system telling us the function and its external entities that serve as the contract between the customer and the developer. It shows system requirements and use to communicate with the system's behavior to the customer and/or the end-user.

System's functionalities are defined by use cases that show specific flow of events. The description of the use case defines the occurrence in the system. Each use case shows a task performed and the collected use cases constitute all the ways in using the system.

The principal symbol representing an actor in UML is shown below:



Figure 4.1 UML representation of an actor.

Determining the actors means establishing the limits of the system so to understand the purpose and its extent. Stakeholders who are directly involved in the system are known as actors. Below are the questions needed to identify the actors:

1. Who will supply, use, or remove information?
2. Who will use this functionality?
3. Who is interested in a certain requirement?
4. Where in the organization is the system used?
5. Who will support and maintain the system?
6. What are the system's external resources?
7. What other systems will need to interact with this one?

After identifying an actor a brief description of it should be given including information about:

1. What or who the actor represents.
2. Why the actor is needed.
3. What interests the actor has in the system.

Understand stakeholder needs

Collect information from the stakeholders to know their needs. The collected information are primarily used as input in defining the high-level features of the system. The purpose of this workflow detail is to elicit and collect information from the stakeholders of the project in order to understand what their needs are. The collected stakeholder requests can be regarded as a "wish list" that will be used as primary input to defining the high-level features of our system, as described in the Vision. This activity is mainly performed during iterations in the inception and elaboration phases.

The key activity of this workflow is to elicit stakeholder requests using such input as business rules, enhancement requests, interviews and requirements workshops. The primary outputs are collections of prioritized features and their critical attributes.

Find use cases

Use case defines the functionality of the system. Use case modeling is a technique in capturing the functional requirements of the system. It describes the occurrences in the system when an actor interacts to it to execute the use case and it specifies the behavior of the system and its subsystems. Each use case represents a flow of events. The flow of events of a use case carries the most important information derived from use-case modelling.

Use cases are graphically represented in oval form.

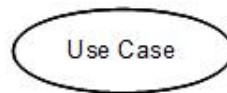


Figure 4.2 UML representation of a use case.

Following questions can be useful when identifying use cases:

1. For each actor you have identified, what are the tasks in which the system would be involved?
2. Does the actor need to be informed about certain occurrences in the system?
3. Will the actor need to inform the system about sudden, external changes?
4. Does the system supply the business with the correct behavior?
5. Can all features be performed by the use cases you have identified?
6. What use cases will support and maintain the system?
7. What information must be modified or created in the system?

Use cases that are often overlooked, since they do not represent what typically are the primary functions of the system, can be of the following kind:

1. System start and stop.
2. Maintenance of the system. For example, adding new users and setting up user profiles.
3. Maintenance of data stored in the system. For example, the system is constructed to work in parallel with a legacy system, and data needs to be synchronized between the two.
4. Functionality needed to modify behavior in the system. An example would be functionality for creating new reports.

Diagrams with actors, use cases, and relationships among them are called use-case diagrams and illustrate relationships in the use-case model.

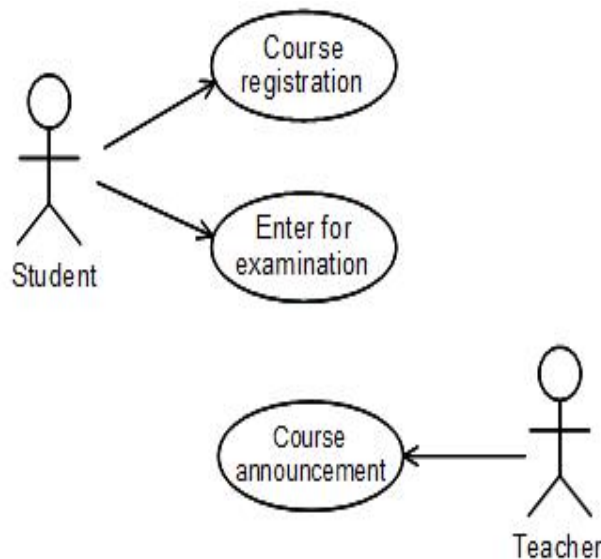


Figure 4.3 Use case diagram in UML.

A use-case model is a model of the system's intended functions and its surroundings and serves as a contract between the customer and the developers. It describes a system's requirements in terms of use cases.

Manage dependencies

The introduction of notion of requirement types helps separate the different levels of abstraction and purposes of the requirements. A sudden change in the requirements will naturally impact the models. Regular reviews, along with updates to the attributes and dependencies, should be done whenever the requirements specifications are updated.

Details of use cases

Use case defines the functionality of the system and represents the specific flow of events. It describes what happens in the system when an actor reacts in an event. Thus, the use case does not define performs internally, collaborating the objects.

An instance in the use case corresponds to the specific flow of an event, invoked by an actor, executing a sequence of events among set of objects known as realization.

The most important information derived from a use-case model is contained in the flow of events. Below are the contents of the flow of events guidelines:

1. Describe how the use case starts and ends
2. Describe what data is exchanged between the actor and the use case
3. Describe the flow of events, not only the functionality. To enforce this, start every action with "When the actor ... "
4. Describe only the events that belong to the use case, and not what happens in other use cases or outside of the system
5. Detail the flow of events, all "what" should be answered. Remember that test designers are to use this text to identify test cases.

Basic flow and alternative flow are the two main parts of the flow of events. The basic flow covers the normal behavior in the execution of the use-case, while the alternative flow includes both optional and exceptional character in relation to the natural behavior and its variations. Both the basic and alternative flow must be structured into steps or sub-flows. A sub-flow is a segment of behavior that has a clear purpose and is atomic that either or neither all nor none of the actions is done.

Extend relationship between use cases

Extend-relationship is an optional component of the use-case. It is used to factor out an additional use-case to simplify the structure of the base.

An extend-relationship is a relationship from an extension use case to a base use case, specifying how the behavior defined for the extension use case can be inserted into the behavior defined for the base use case. It is implicitly inserted in the sense that the extension is not shown in the base use case. The extensions can be used for several purposes:

1. To show that a part of a use case is optional, or potentially optional, system behavior.
2. To show that a sub-flow is executed only under certain conditions, such as triggering an alarm.
3. To show that there may be a set of behavior segments of which one or several may be inserted at an extension point in a base use case. The behavior segments that are inserted will depend on the interaction with the actors during the execution of the base use case.

The execution of the extension is conditional to what is happening in the execution of the base use-case. While the base use-case does not control the execution of the extension. Further, the condition therein is only described in the extend relationship. Extensions modify the execution of the base use-case as it defines a modular framework into which extensions can be added but does not have any visibility of the specific extensions.

Example for application of extend relationship.

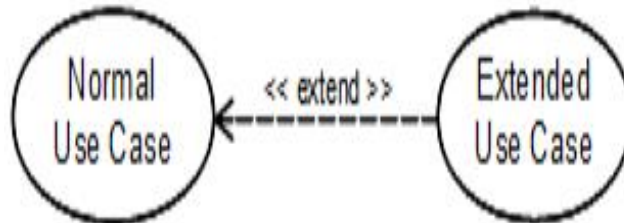


Figure 4.4 Representation of extend relationship.

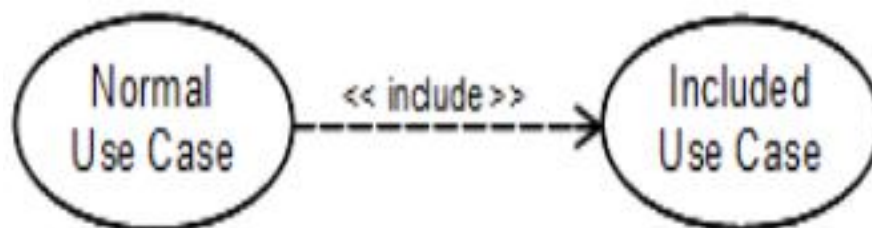
Include relationship between use cases

Include-relationship represents a function at which the use case only depends on the result but not the method used to produce the result.

The include-relationship is always an abstract that connects a base use case to an inclusion use case. It describes a behavior segment that is inserted into a use-case instance that is executing the base use case. The base use case has control of the relationship to the inclusion and can depend on the result of performing the inclusion, but neither the base nor the inclusion may access each other's attributes. The inclusion is in this sense encapsulated and represents behavior that can be reused in different base use cases.

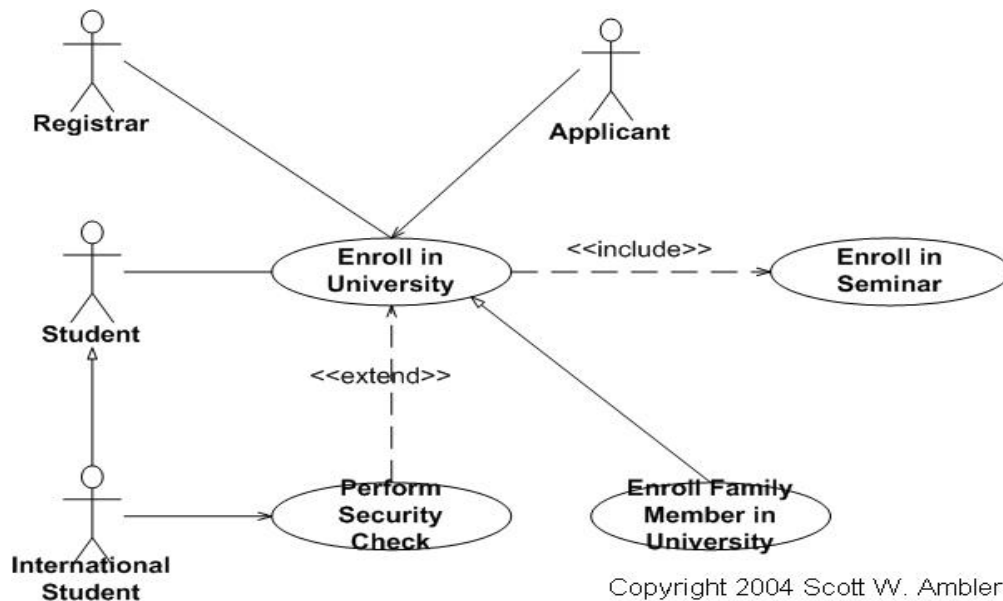
Reasons for using the include-relationship are as follows:

1. Factor out behavior from the base use case that is not necessary for the understanding of the primary purpose of the use case, only the result of it is important.
2. Factor out behavior that is in common for two or more use cases.



http://moodle.autolab.uni-pannon.hu/Mecha_tananyag/szoftverfejlesztési_folyamatok_angol/ch07.html

Figure 4.5 Representation of include relationship.



<http://agilemodeling.com/essays/useCaseReuse.htm>

Figure 4.6 Example of a Use-Case Diagram with extend and include.

Software Development Life Cycle (SDLC)

Software development life cycle is a pictogram that represents the methods required in creating a software. It captures the methods the software will undertake. In other words, it is mapping of various activities from its interception until its terminal.

Different life cycle models may plan the necessary development activities to phases in different ways. Thus, no element which life cycle model is followed; the essential activities are contained in all life cycle models though the action may be carried out in distinct orders in different life cycle models. During any life cycle stage, more than one activity may also be carried out.

Need of SDLC

The development team must determine a suitable life cycle model for a particular plan and then observe to it.

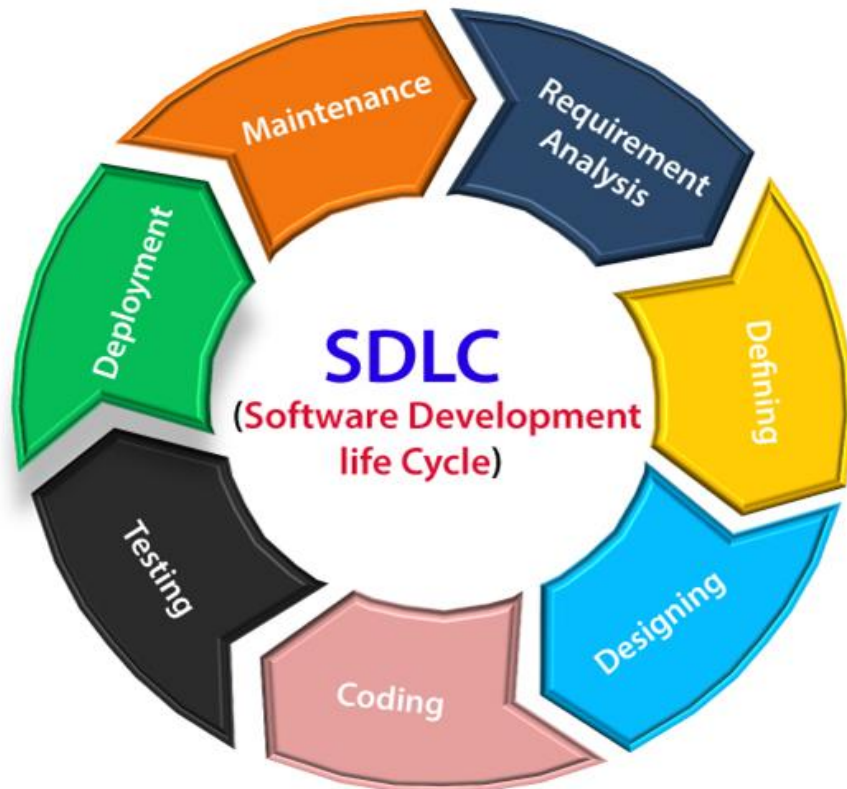
Without using an exact life cycle model, the development of a software product would not be in a systematic and disciplined manner. When a team is developing a software product, there must be a clear understanding among team representative about when and what to do. Otherwise, it would point to chaos and project failure. This problem can be defined by using an example. Suppose a software development issue is divided into various parts and the parts are assigned to the team members. From then on, suppose the team representative is allowed the freedom to develop the roles assigned to them in whatever way they like. It is possible that one representative might start writing the code for his part, another might choose to prepare the test documents first, and some other engineer might begin with the design phase of the roles assigned to him. This would be one of the perfect methods for project failure.

A software life cycle model describes entry and exit criteria for each phase. A phase can begin only if its stage-entry criteria have been fulfilled. So, without a software life cycle

model, the entry and exit criteria for a stage cannot be recognized. Without software life cycle models, it becomes tough for software project managers to monitor the progress of the project.

SDLC Cycle

SDLC Cycle represents the process of developing software. SDLC framework includes the following steps:



Source: <https://www.javatpoint.com/software-engineering-software-development-life-cycle>

Figure 4.7 Software Development Life Cycle (SDLC)

The stages of SDLC are as follows:

Stage 1: Planning and requirement analysis

Requirement Analysis is the most important and necessary stage in SDLC.

The senior members of the team perform it with inputs from all the stakeholders and domain experts or SMEs in the industry. Planning for the quality assurance requirements and identifications of the risks associated with the projects is also done at this stage.

Business analyst and Project organizer set up a meeting with the client to gather all the data like what the customer wants to build, who will be the end user, what is the objective of the product. Before creating a product, a core understanding or knowledge of the product is very necessary.

For Example, *A client wants to have an application which concerns money transactions. In this method, the requirement has to be precise like what kind of operations will be done, how it will be done, in which currency it will be done, etc.*

Once the required function is done, an analysis is complete with auditing the feasibility of the growth of a product. In case of any ambiguity, a signal is set up for further discussion. Once the requirement is understood, the SRS (Software Requirement Specification) document is created. The developers should thoroughly follow this document and also should be reviewed by the customer for future reference.

Stage 2: Defining Requirements

Once the requirement analysis is done, the next stage is to certainly represent and document the software requirements and get them accepted from the project stakeholders.

This is accomplished through "SRS"- Software Requirement Specification document which contains all the product requirements to be constructed and developed during the project life cycle.

Stage 3: Designing the Software

The next phase is about to bring down all the knowledge of requirements, analysis, and design of the software project. This phase is the product of the last two, like inputs from the customer and requirement gathering.

Stage 4: Developing the project

In this phase of SDLC, the actual development begins, and the programming is built. The implementation of design begins concerning writing code. Developers have to follow the coding guidelines described by their management and programming tools like compilers, interpreters, debuggers, etc. are used to develop and implement the code.

Stage 5: Testing

After the code is generated, it is tested against the requirements to make sure that the products are solving the needs addressed and gathered during the requirements stage.

During this stage, unit testing, integration testing, system testing, acceptance testing is done.

Stage 6: Deployment

Once the software is certified, and no bugs or errors are stated, then it is deployed.

Then based on the assessment, the software may be released as it is or with suggested enhancement in the object segment.

After the software is deployed, then its maintenance begins.

Stage 7: Maintenance

Once when the client starts using the developed systems, then the real issues come up and requirements to be solved from time to time.

This procedure where the care is taken for the developed product is known as maintenance.

<https://www.javatpoint.com/software-engineering-software-development-life-cycle>

SDLC Models

Software Development life cycle (SDLC) is a spiritual model used in project management that defines the stages include in an information system development project, from an initial feasibility study to the maintenance of the completed application.

There are different software development life cycle models specify and design, which are followed during the software development phase. These models are also called "**Software Development Process Models.**" Each process model follows a series of phase unique to its type to ensure success in the step of software development.

Here, are some important phases of SDLC life cycle:

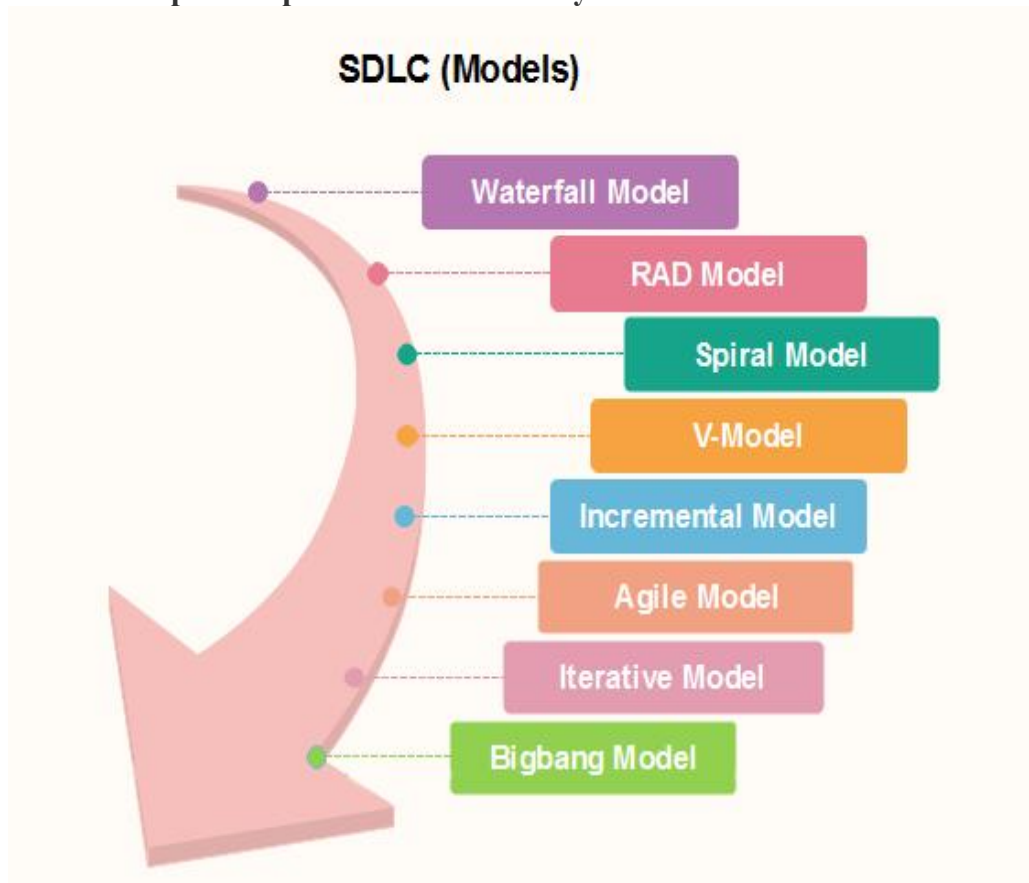


Figure 4.8 SDLC Models

Waterfall Model

The waterfall is a universally accepted SDLC model. In this method, the whole process of software development is divided into various phases.

The waterfall model is a continuous software development model in which development is seen as flowing steadily downwards (like a waterfall) through the steps of requirements analysis, design, implementation, testing (validation), integration, and maintenance.

Linear ordering of activities has some significant consequences. First, to identify the end of a phase and the beginning of the next, some certification techniques have to be employed at the end of each step. Some verification and validation usually do this mean that will ensure that the output of the stage is consistent with its input (which is the output of the previous step), and that the output of the stage is consistent with the overall requirements of the system.

RAD Model

RAD or Rapid Application Development process is an adoption of the waterfall model; it targets developing software in a short period. The RAD model is based on the concept that a better system can be developed in lesser time by using focus groups to gather system requirements.

- Business Modeling
- Data Modeling
- Process Modeling
- Application Generation
- Testing and Turnover

Spiral Model

The spiral model is a **risk-driven process model**. This SDLC model helps the group to adopt elements of one or more process models like a waterfall, incremental, waterfall, etc. The spiral technique is a combination of rapid prototyping and concurrency in design and development activities.

Each cycle in the spiral begins with the identification of objectives for that cycle, the different alternatives that are possible for achieving the goals, and the constraints that exist. This is the first quadrant of the cycle (upper-left quadrant).

The next step in the cycle is to evaluate these different alternatives based on the objectives and constraints. The focus of evaluation in this step is based on the risk perception for the project.

The next step is to develop strategies that solve uncertainties and risks. This step may involve activities such as benchmarking, simulation, and prototyping.

V-Model

In this type of SDLC model testing and the development, the step is planned in parallel. So, there are verification phases on the side and the validation phase on the other side. V-Model joins by Coding phase.

Incremental Model

The incremental model is not a separate model. It is necessarily a series of waterfall cycles. The requirements are divided into groups at the start of the project. For each group, the SDLC model is followed to develop software. The SDLC process is repeated, with each release adding more functionality until all requirements are met. In this method, each cycle act as the maintenance phase for the previous software release. Modification to the incremental model allows development cycles to overlap. After that subsequent cycle may begin before the previous cycle is complete.

Agile Model

Agile methodology is a practice which promotes continues interaction of development and testing during the SDLC process of any project. In the Agile method, the entire project is

divided into small incremental builds. All of these builds are provided in iterations, and each iteration lasts from one to three weeks.

Any agile software phase is characterized in a manner that addresses several key assumptions about the bulk of software projects:

1. It is difficult to think in advance which software requirements will persist and which will change. It is equally difficult to predict how user priorities will change as the project proceeds.
2. For many types of software, design and development are interleaved. That is, both activities should be performed in tandem so that design models are proven as they are created. It is difficult to think about how much design is necessary before construction is used to test the configuration.
3. Analysis, design, development, and testing are not as predictable (from a planning point of view) as we might like.

Iterative Model

It is a particular implementation of a software development life cycle that focuses on an initial, simplified implementation, which then progressively gains more complexity and a broader feature set until the final system is complete. In short, iterative development is a way of breaking down the software development of a large application into smaller pieces.

Big bang model

Big bang model is focusing on all types of resources in software development and coding, with no or very little planning. The requirements are understood and implemented when they come.

This model works best for small projects with smaller size development team which are working together. It is also useful for academic software development projects. It is an ideal model where requirements are either unknown or final release date is not given.

Prototype Model

The prototyping model starts with the requirements gathering. The developer and the user meet and define the purpose of the software, identify the needs, etc.

A '**quick design**' is then created. This design focuses on those aspects of the software that will be visible to the user. It then leads to the development of a prototype. The customer then checks the prototype, and any modifications or changes that are needed are made to the prototype.

Looping takes place in this step, and better versions of the prototype are created. These are continuously shown to the user so that any new changes can be updated in the prototype. This process continues until the customer is satisfied with the system. Once a user is satisfied, the prototype is converted to the actual system with all considerations for quality and security. Source: <https://www.javatpoint.com/software-engineering-sdlc-models>

In general, writing a capstone project SDLC, each phase must contain the procedure or processes undertaken in the software development.

For example, the first phase in the software development would talk about planning and requirements analysis.

The guiding questions may be necessary in writing the procedural statements:

1. How was the problem determined?
2. What process is used to determine the problem?
3. Was there a feasibility study conducted?
4. Were there series of interview made? If there were interviews done, state the manner and how the interview was conducted.
5. How were the data being gathered? (i.e. through online platform, physical instruments were distributed, etc.)
6. How are the data gathered being analyzed? Were there statistical tools used?

Each statement written in the phases of the SDLC must conform with all the procedures/processing used.

4.3 Requirement Documentation

Before the commencement of software design, requirement documentation must be provided. This is a legal requirement in a real-life situation. These establishes the basis for agreement between the customer/client and the developers/programmers on what software product is to do.

Under this section, all software features are enumerated in detail by providing story board showing how the software would look like if the same was already designed and coded. Requirements documents states what the software will do. It does not state how the software will do it. This is because requirement documentation is written both for the programmers and the client. Hence, customers or clients can understand the requirements documentation if the programmer/proponent will provide WHAT the software will do. Therefore, the requirements documentation does not include database schemas, normalization of tables, system architectures, descriptions of communication layers- in short, no statements of design of any sort. The answer to the question “HOW the software will do” or statement of design shall be addressed under results and discussion.

Berezon (1994) provides a good example; it is a requirement for a word processing application to be able to open an existing file. It is a design issue whether to build a customized file selection tool or use a platform- standard file selection tool. The requirements document brings the following additional benefits:

- The customers can see early on if their needs will be met
- The developers can estimate the effort involved in creating the application
- The development project leader has a basis for a project plan
- The quality assurance people have a basis for testing the application

Obviously, the proponent will need data gathering tools in preparing his requirements analysis and documentation.

Example: Source: (Laviña, C., et. Al., Outcomes-Based Practical Guide to Thesis and Capstone Project Writing in Computing, Mindshapers Co., Inc. 2016)

The following is a part of the Requirement Documentation of the capstone project entitles” *An Online Payroll System with R3 Report Generator of SSS and Employee Monitoring for Edcar Manpower Services.*”

REQUIREMENT DOCUMENTATION

Project In- Scope and Out- Scope

In- scope

● Account Module

This module will utilize an employee management that makes adding, editing and deleting of employees. The system also shows details of the employee such as “active”, and “inactive” among others.

● Payroll Module

This module will create a reliable and accurate payroll system that will generate reports based on different primary keys such as employees, dates, etc. the report may include at the option of the users the following: SSS contributions, Philhealth, Union dues, PAGIBIG, Loan payment, among others.

● File Archive Mode

This module will provide a data file archive that allows backing up files and data records that can be exported from database.

Requirement Engineering

Requirements engineering (RE) refers to the process of defining, documenting, and maintaining requirements in the engineering design process. Requirement engineering provides the appropriate mechanism to understand what the customer desires, analyzing the need, and assessing feasibility, negotiating a reasonable solution, specifying the solution clearly, validating the specifications and managing the requirements as they are transformed into a working system. Thus, requirement engineering is the disciplined application of proven principles, methods, tools, and notation to describe a proposed system's intended behavior and its associated constraints.

Requirement Engineering Process is a four-step process, which includes –

1. Feasibility Study
2. Requirement Elicitation and Analysis
3. Software Requirement Specification
4. Software Requirement Validation
5. Software Requirement Management
6. Figure 4.8 SDLC Models

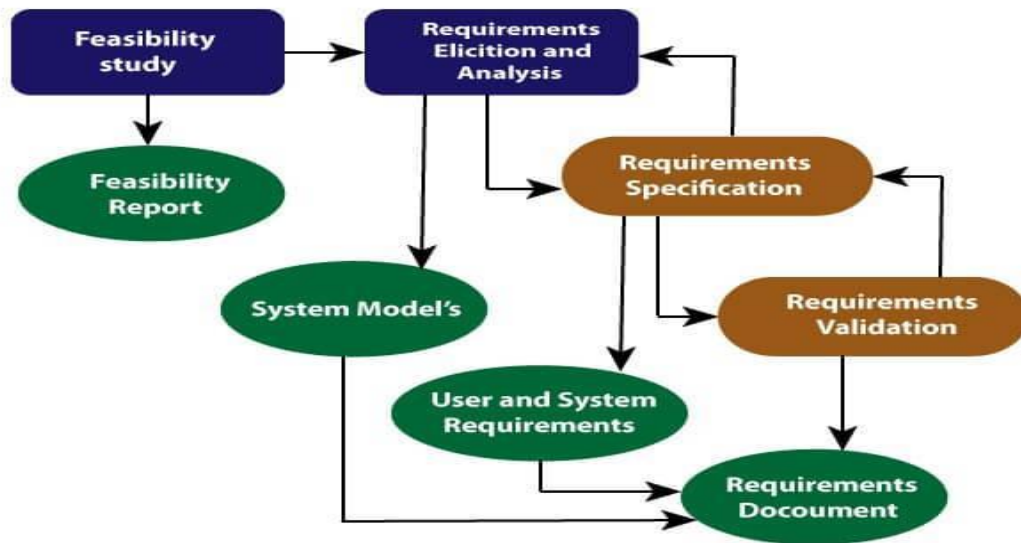


Figure 4.9 Requirement Engineering Process

1. Feasibility Study:

The objective behind the feasibility study is to create the reasons for developing the software that is acceptable to users, flexible to change and conformable to established standards.

Types of Feasibility:

1. **Technical Feasibility** - Technical feasibility evaluates the current technologies, which are needed to accomplish customer requirements within the time and budget.
2. **Operational Feasibility** - Operational feasibility assesses the range in which the required software performs a series of levels to solve business problems and customer requirements.
3. **Economic Feasibility** - Economic feasibility decides whether the necessary software can generate financial profits for an organization.

2. Requirement Elicitation and Analysis:

This is also known as the **gathering of requirements**. Here, requirements are identified with the help of customers and existing systems processes, if available.

Analysis of requirements starts with requirement elicitation. The requirements are analyzed to identify inconsistencies, defects, omission, etc. We describe requirements in terms of relationships and also resolve conflicts if any.

Problems of Elicitation and Analysis

- Getting all, and only, the right people involved.
- Stakeholders often don't know what they want
- Stakeholders' express requirements in their terms.
- Stakeholders may have conflicting requirements.
- Requirement change during the analysis process.
- Organizational and political factors may influence system requirements.

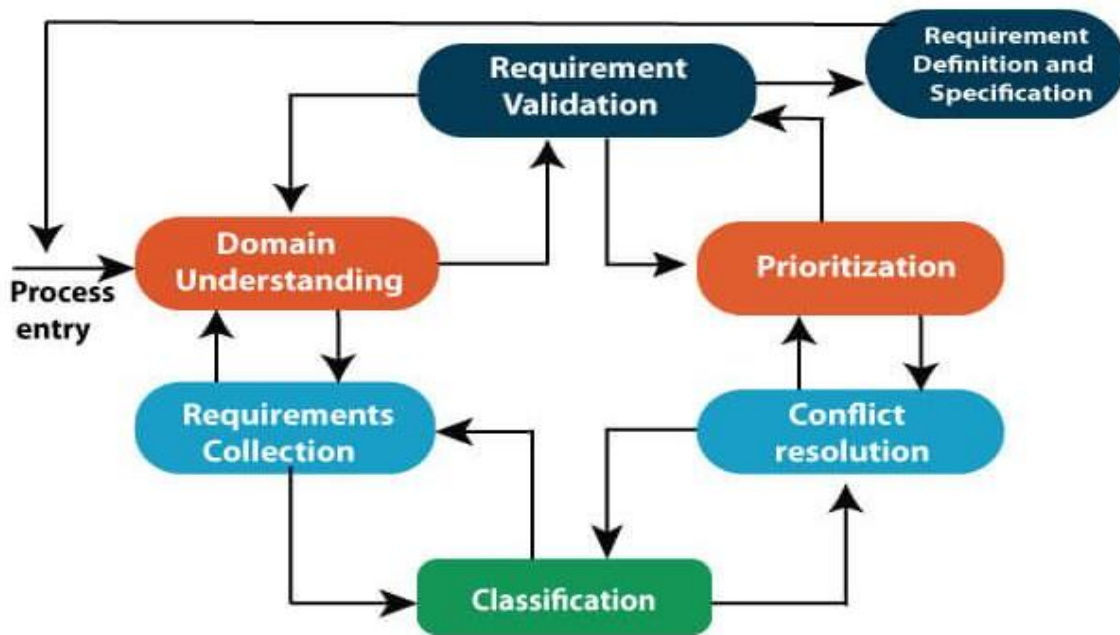


Figure 4.10 Elicitation and Analysis Process

3. Software Requirement Specification:

Software requirement specification is a kind of document which is created by a software analyst after the requirements collected from the various sources - the requirement received by the customer written in ordinary language. It is the job of the analyst to write the requirement in technical language so that they can be understood and beneficial by the development team.

The models used at this stage include ER diagrams, data flow diagrams (DFDs), function decomposition diagrams (FDDs), data dictionaries, etc.

- **Data Flow Diagrams:** Data Flow Diagrams (DFDs) are used widely for modeling the requirements. DFD shows the flow of data through a system. The system may be a company, an organization, a set of procedures, a computer hardware system, a software system, or any combination of the preceding. The DFD is also known as a data flow graph or bubble chart.
- **Data Dictionaries:** Data Dictionaries are simply repositories to store information about all data items defined in DFDs. At the requirements stage, the data dictionary should at least define customer data items, to ensure that the customer and developers use the same definition and terminologies.
- **Entity-Relationship Diagrams:** Another tool for requirement specification is the entity-relationship diagram, often called an "*E-R diagram*." It is a detailed logical representation of the data for the organization and uses three main constructs i.e. data entities, relationships, and their associated attributes.

4. Software Requirement Validation:

After requirement specifications developed, the requirements discussed in this document are validated. The user might demand illegal, impossible solution or experts may misinterpret the needs. Requirements can be the check against the following conditions -

- If they can practically implement

- If they are correct and as per the functionality and specially of software
- If there are any ambiguities
- If they are full
- If they can describe

Requirements Validation Techniques

- **Requirements reviews/inspections:** systematic manual analysis of the requirements.
- **Prototyping:** Using an executable model of the system to check requirements.
- **Test-case generation:** Developing tests for requirements to check testability.
- **Automated consistency analysis:** checking for the consistency of structured requirements descriptions.

Software Requirement Management:

Requirement management is the process of managing changing requirements during the requirements engineering process and system development.

New requirements emerge during the process as business needs a change, and a better understanding of the system is developed.

The priority of requirements from different viewpoints changes during development process.

The business and technical environment of the system changes during the development.

Prerequisite of Software requirements

Collection of software requirements is the basis of the entire software development project. Hence, they should be clear, correct, and well-defined.

A complete Software Requirement Specifications should be:

- Clear
- Correct
- Consistent
- Coherent
- Comprehensible
- Modifiable
- Verifiable
- Prioritized
- Unambiguous
- Traceable
- Credible source

Software Requirements: Largely software requirements must be categorized into two categories:

1. **Functional Requirements:** Functional requirements define a function that a system or system element must be qualified to perform and must be documented in different forms. The functional requirements are describing the behavior of the system as it correlates to the system's functionality.

2. **Non-functional Requirements:** This can be the necessities that specify the criteria that can be used to decide the operation instead of specific behaviors of the system.

Non-functional requirements are divided into two main categories:

1. **Execution qualities** like security and usability, which are observable at run time.
2. **Evolution qualities** like testability, maintainability, extensibility, and scalability that embodied in the static structure of the software system.

Source: <https://www.javatpoint.com/software-engineering-requirement-engineering>

Waterfall model

Winston Royce introduced the Waterfall Model in 1970. This model has five phases: Requirement's analysis and specification, design, implementation, and unit testing, integration and system testing, and operation and maintenance. The steps always follow in this order and do not overlap. The developer must complete every phase before the next phase begins. This model is named "**Waterfall Model**", because its diagrammatic representation resembles a cascade of waterfalls.

1. Requirements analysis and specification phase: The aim of this phase is to understand the exact requirements of the customer and to document them properly. Both the customer and the software developer work together so as to document all the functions, performance, and interfacing requirement of the software. It describes the "what" of the system to be produced and not "how." In this phase, a large document called **Software Requirement Specification (SRS)** document is created which contained a detailed description of what the system will do in the common language.

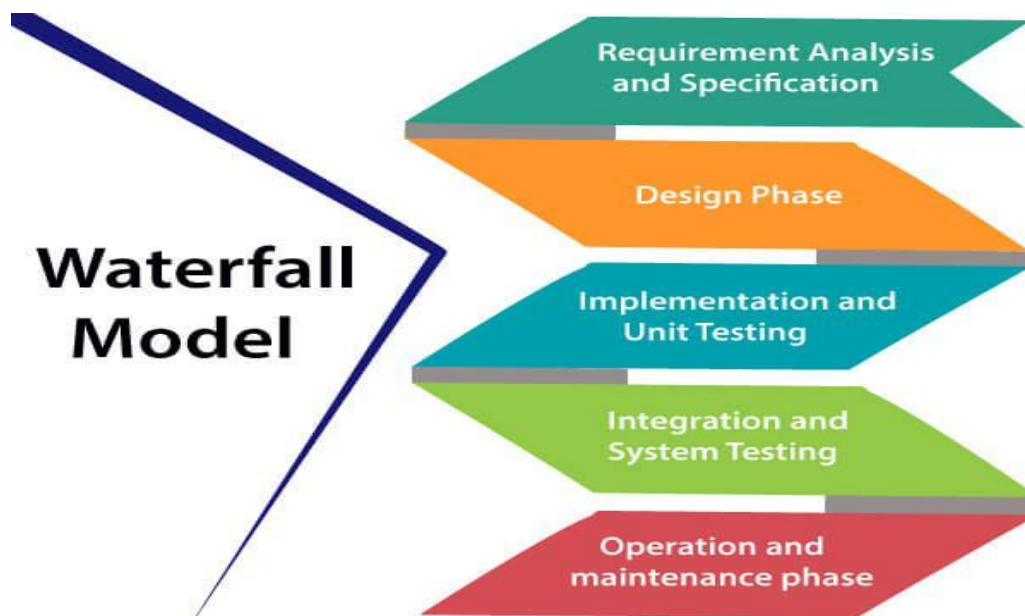


Figure 4.11 Waterfall Model

2. Design Phase: This phase aims to transform the requirements gathered in the SRS into a suitable form which permits further coding in a programming language. It defines the overall

software architecture together with high level and detailed design. All this work is documented as a Software Design Document (SDD).

3. Implementation and unit testing: During this phase, design is implemented. If the SDD is complete, the implementation or coding phase proceeds smoothly, because all the information needed by software developers is contained in the SDD.

During testing, the code is thoroughly examined and modified. Small modules are tested in isolation initially. After that these modules are tested by writing some overhead code to check the interaction between these modules and the flow of intermediate output.

4. Integration and System Testing: This phase is highly crucial as the quality of the end product is determined by the effectiveness of the testing carried out. The better output will lead to satisfied customers, lower maintenance costs, and accurate results. Unit testing determines the efficiency of individual modules. However, in this phase, the modules are tested for their interactions with each other and with the system.

5. Operation and maintenance phase: Maintenance is the task performed by every user once the software has been delivered to the customer, installed, and operational.

When to use SDLC Waterfall Model?

Some Circumstances where the use of the Waterfall model is most suited are:

- When the requirements are constant and not changed regularly.
- A project is short
- The situation is calm
- Where the tools and technology used is consistent and is not changing
- When resources are well prepared and are available to use.

Advantages of Waterfall model

- This model is simple to implement also the number of resources that are required for it is minimal.
- The requirements are simple and explicitly declared; they remain unchanged during the entire project development.
- The start and end points for each phase is fixed, which makes it easy to cover progress.
- The release date for the complete product, as well as its final cost, can be determined before development.
- It gives easy to control and clarity for the customer due to a strict reporting system.

Disadvantages of Waterfall model

- In this model, the risk factor is higher, so this model is not suitable for more significant and complex projects.
- This model cannot accept the changes in requirements during development.
- It becomes tough to go back to the phase. For example, if the application has now shifted to the coding phase, and there is a change in requirement, It becomes tough to go back and change it.

- Since the testing done at a later stage, it does not allow identifying the challenges and risks in the earlier phase, so the risk reduction strategy is difficult to prepare.

Source: <https://www.javatpoint.com/software-engineering-waterfall-model>

RAD (Rapid Application Development) Model

RAD is a linear sequential software development process model that emphasizes a concise development cycle using an element based construction approach. If the requirements are well understood and described, and the project scope is a constraint, the RAD process enables a development team to create a fully functional system within a concise time period.

RAD (Rapid Application Development) is a concept that products can be developed faster and of higher quality through:

- Gathering requirements using workshops or focus groups
- Prototyping and early, reiterative user testing of designs
- The re-use of software components
- A rigidly paced schedule that refers design improvements to the next product version
- Less formality in reviews and other team communication

Fig: RAD Model

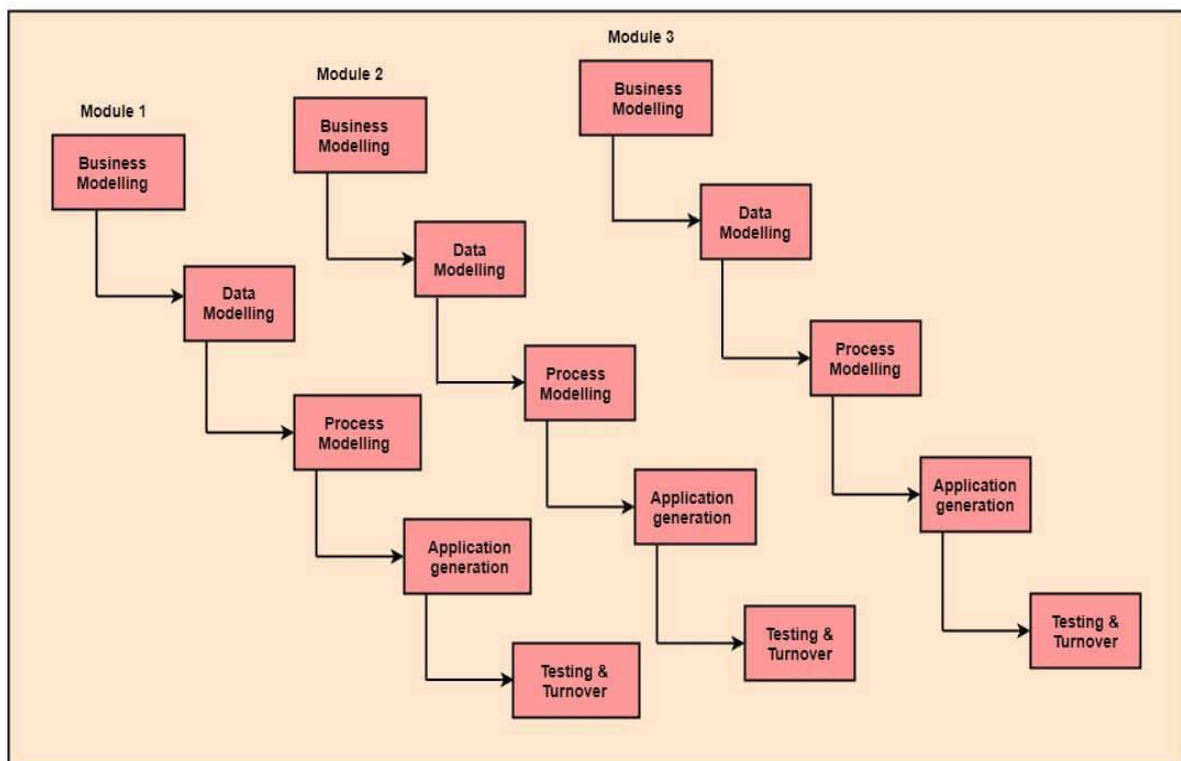


Figure 4.12 RAD Model

The various phases of RAD are as follows:

1.Business Modelling: The information flow among business functions is defined by answering questions like what data drives the business process, what data is generated, who generates it, where does the information go, who process it and so on.

- 2. Data Modelling:** The data collected from business modeling is refined into a set of data objects (entities) that are needed to support the business. The attributes (character of each entity) are identified, and the relation between these data objects (entities) is defined.
- 3. Process Modelling:** The information object defined in the data modeling phase are transformed to achieve the data flow necessary to implement a business function. Processing descriptions are created for adding, modifying, deleting, or retrieving a data object.
- 4. Application Generation:** Automated tools are used to facilitate construction of the software; even they use the 4th GL techniques.
- 5. Testing & Turnover:** Many of the programming components have already been tested since RAD emphasis reuse. This reduces the overall testing time. But the new part must be tested, and all interfaces must be fully exercised.

When to use RAD Model?

- When the system should need to create the project that modularizes in a short span time (2-3 months).
- When the requirements are well-known.
- When the technical risk is limited.
- When there's a necessity to make a system, which modularized in 2-3 months of period.
- It should be used only if the budget allows the use of automatic code generating tools.

Advantage of RAD Model

- This model is flexible for change.
- In this model, changes are adoptable.
- Each phase in RAD brings highest priority functionality to the customer.
- It reduced development time.
- It increases the reusability of features.

Disadvantage of RAD Model

- It required highly skilled designers.
- All application is not compatible with RAD.
- For smaller projects, we cannot use the RAD model.
- On the high technical risk, it's not suitable.
- Required user involvement.

Source: <https://www.javatpoint.com/software-engineering-rapid-application-development-model>

Spiral Model

The spiral model, initially proposed by Boehm, is an evolutionary software process model that couples the iterative feature of prototyping with the controlled and systematic aspects of the linear sequential model. It implements the potential for rapid development of new versions of the software. Using the spiral model, the software is developed in a series of incremental releases. During the early iterations, the additional release may be a paper model or prototype. During later iterations, more and more complete versions of the engineered system are produced.

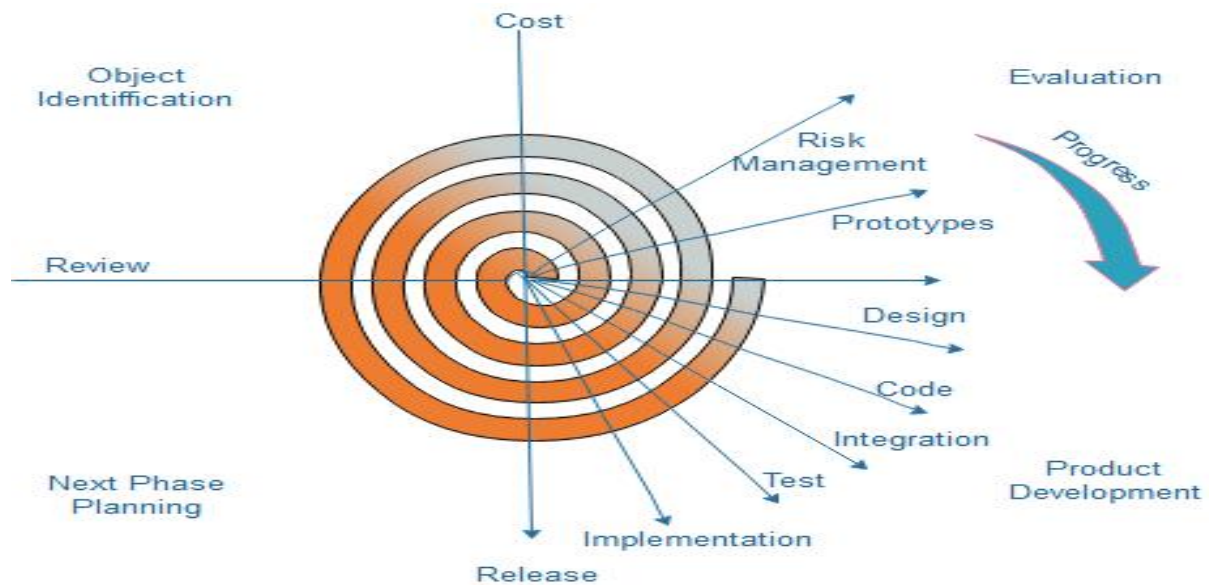


Figure 4.13 Spiral Model

Each cycle in the spiral is divided into four parts:

Objective setting: Each cycle in the spiral starts with the identification of purpose for that cycle, the various alternatives that are possible for achieving the targets, and the constraints that exists.

Risk Assessment and reduction: The next phase in the cycle is to calculate these various alternatives based on the goals and constraints. The focus of evaluation in this stage is located on the risk perception for the project.

Development and validation: The next phase is to develop strategies that resolve uncertainties and risks. This process may include activities such as benchmarking, simulation, and prototyping.

Planning: Finally, the next step is planned. The project is reviewed, and a choice made whether to continue with a further period of the spiral. If it is determined to keep, plans are drawn up for the next step of the project.

The development phase depends on the remaining risks. For example, if performance or user-interface risks are treated more essential than the program development risks, the next phase may be an evolutionary development that includes developing a more detailed prototype for solving the risks.

The **risk-driven** feature of the spiral model allows it to accommodate any mixture of a specification-oriented, prototype-oriented, simulation-oriented, or another type of approach. An essential element of the model is that each period of the spiral is completed by a review that includes all the products developed during that cycle, including plans for the next cycle. The spiral model works for development as well as enhancement projects.

When to use Spiral Model?

- When deliverance is required to be frequent.

- When the project is large
- When requirements are unclear and complex
- When changes may require at any time
- Large and high budget projects

Advantages

- High amount of risk analysis
- Useful for large and mission-critical projects.

Disadvantages

- Can be a costly model to use.
- Risk analysis needed highly particular expertise
- Doesn't work well for smaller projects.

Source: <https://www.javatpoint.com/software-engineering-spiral-model>

V-Model

V-Model also referred to as the Verification and Validation Model. In this, each phase of SDLC must complete before the next phase starts. It follows a sequential design process same as the waterfall model. Testing of the device is planned in parallel with a corresponding stage of development.

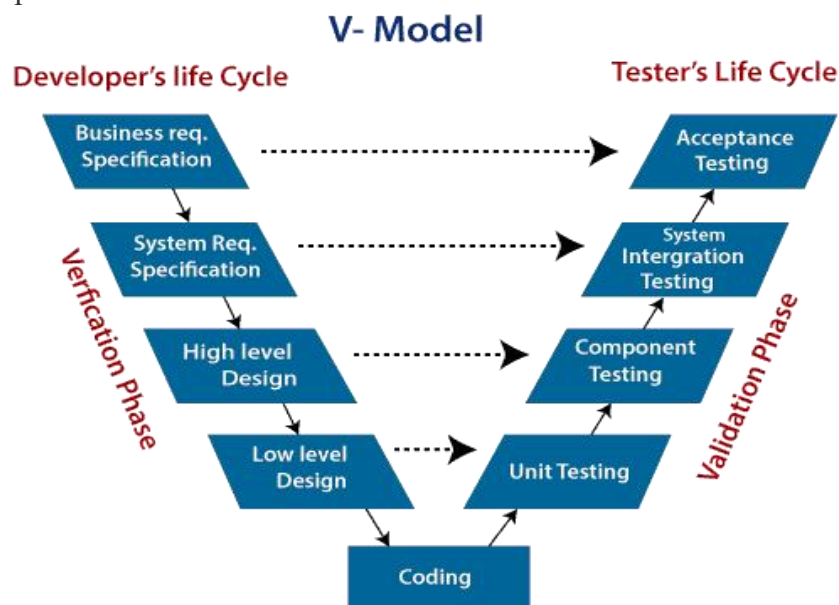


Figure 4.14 V-Model

Verification: It involves a static analysis method (review) done without executing code. It is the process of evaluation of the product development process to find whether specified requirements meet.

Validation: It involves dynamic analysis method (functional, non-functional), testing is done by executing code. Validation is the process to classify the software after the completion of

the development process to determine whether the software meets the customer expectations and requirements.

So V-Model contains Verification phases on one side of the Validation phases on the other side. Verification and Validation process is joined by coding phase in V-shape. Thus it is known as V-Model.

There are the various phases of Verification Phase of V-model:

Business requirement analysis: This is the first step where product requirements understood from the customer's side. This phase contains detailed communication to understand customer's expectations and exact requirements.

System Design: In this stage system engineers analyze and interpret the business of the proposed system by studying the user requirements document.

Architecture Design: The baseline in selecting the architecture is that it should understand all which typically consists of the list of modules, brief functionality of each module, their interface relationships, dependencies, database tables, architecture diagrams, technology detail, etc. The integration testing model is carried out in a particular phase.

Module Design: In the module design phase, the system breaks down into small modules. The detailed design of the modules is specified, which is known as Low-Level Design

Coding Phase: After designing, the coding phase is started. Based on the requirements, a suitable programming language is decided. There are some guidelines and standards for coding. Before checking in the repository, the final build is optimized for better performance, and the code goes through many code reviews to check the performance.

There are the various phases of Validation Phase of V-model:

Unit Testing: In the V-Model, Unit Test Plans (UTPs) are developed during the module design phase. These UTPs are executed to eliminate errors at code level or unit level. A unit is the smallest entity which can independently exist, e.g., a program module. Unit testing verifies that the smallest entity can function correctly when isolated from the rest of the codes/ units.

Integration Testing: Integration Test Plans are developed during the Architectural Design Phase. These tests verify that groups created and tested independently can coexist and communicate among themselves.

System Testing: System Tests Plans are developed during System Design Phase. Unlike Unit and Integration Test Plans, System Tests Plans are composed by the client's business team. System Test ensures that expectations from an application developer are met.

Acceptance Testing: Acceptance testing is related to the business requirement analysis part. It includes testing the software product in user atmosphere. Acceptance tests reveal the compatibility problems with the different systems, which is available within the user atmosphere. It conjointly discovers the non-functional problems like load and performance defects within the real user atmosphere.

When to use V-Model?

- When the requirement is well defined and not ambiguous.
- The V-shaped model should be used for small to medium-sized projects where requirements are clearly defined and fixed.
- The V-shaped model should be chosen when sample technical resources are available with essential technical expertise.

Advantage (Pros) of V-Model:

- Easy to Understand.
- Testing Methods like planning, test designing happens well before coding.
- This saves a lot of time. Hence a higher chance of success over the waterfall model.
- Avoids the downward flow of the defects.
- Works well for small plans where requirements are easily understood.

Disadvantage (Cons) of V-Model:

- Very rigid and least flexible.
- Not a good for a complex project.
- Software is developed during the implementation stage, so no early prototypes of the software are produced.
- If any changes happen in the midway, then the test documents along with the required documents, has to be updated.

Source: <https://www.javatpoint.com/software-engineering-v-model>

Incremental Model

Incremental Model is a process of software development where requirements divided into multiple standalone modules of the software development cycle. In this model, each module goes through the requirements, design, implementation and testing phases. Every subsequent release of the module adds function to the previous release. The process continues until the complete system achieved.

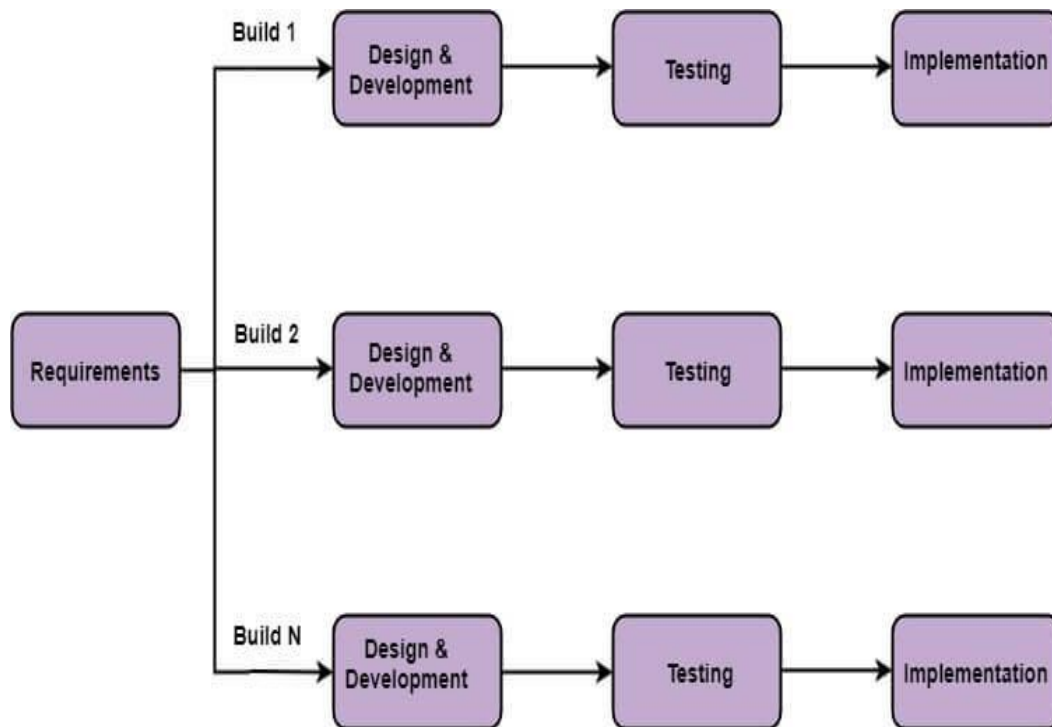


Figure 4.15 Incremental Model

The various phases of incremental model are as follows:

- 1. Requirement analysis:** In the first phase of the incremental model, the product analysis expertise identifies the requirements. And the system functional requirements are understood by the requirement analysis team. To develop the software under the incremental model, this phase performs a crucial role.
- 2. Design & Development:** In this phase of the Incremental model of SDLC, the design of the system functionality and the development method are finished with success. When software develops new practicality, the incremental model uses style and development phase.
- 3. Testing:** In the incremental model, the testing phase checks the performance of each existing function as well as additional functionality. In the testing phase, the various methods are used to test the behavior of each task.
- 4. Implementation:** Implementation phase enables the coding phase of the development system. It involves the final coding that design in the designing and development phase and tests the functionality in the testing phase. After completion of this phase, the number of the product working is enhanced and upgraded up to the final system product

When we use the Incremental Model?

- When the requirements are superior.
- A project has a lengthy development schedule.
- When Software team are not very well skilled or trained.
- When the customer demands a quick release of the product.
- You can develop prioritized requirements first.

Advantage of Incremental Model

- Errors are easy to be recognized.
- Easier to test and debug
- More flexible.
- Simple to manage risk because it handled during its iteration.
- The Client gets important functionality early.

Disadvantage of Incremental Model

- Need for good planning
- Total Cost is high.
- Well defined module interfaces are needed.

Source: <https://www.javatpoint.com/software-engineering-incremental-model>

Agile Model

The meaning of Agile is swift or versatile. “**Agile process model**” refers to a software development approach based on iterative development. Agile methods break tasks into smaller iterations, or parts do not directly involve long term planning. The project scope and requirements are laid down at the beginning of the development process. Plans regarding the number of iterations, the duration and the scope of each iteration are clearly defined in advance.

Each iteration is considered as a short time "frame" in the Agile process model, which typically lasts from one to four weeks. The division of the entire project into smaller parts helps to minimize the project risk and to reduce the overall project delivery time requirements. Each iteration involves a team working through a full software development life cycle including planning, requirements analysis, design, coding, and testing before a working product is demonstrated to the client.

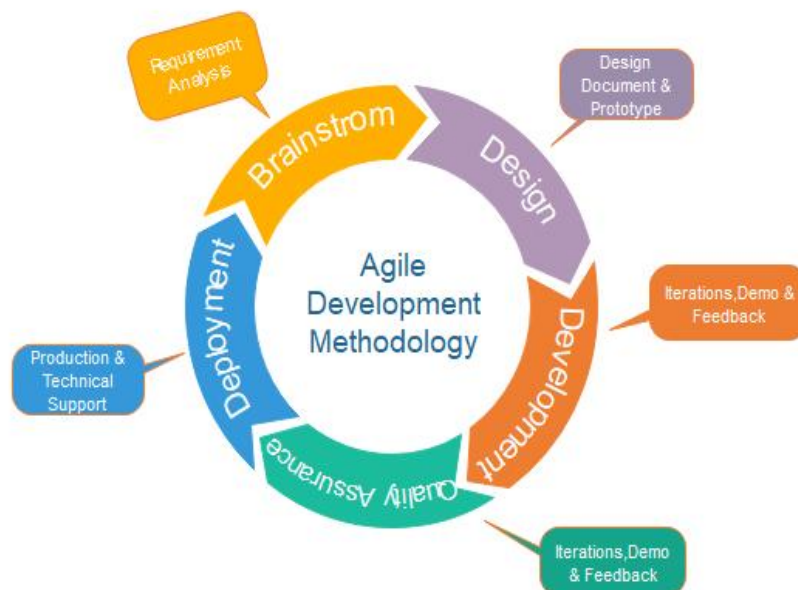


Figure 4.16 Agile Model

Phases of Agile Model:

Following are the phases in the Agile model are as follows:

1. Requirements gathering
2. Design the requirements
3. Construction/ iteration
4. Testing/ Quality assurance
5. Deployment
6. Feedback

- 1. Requirements gathering:** In this phase, you must define the requirements. You should explain business opportunities and plan the time and effort needed to build the project. Based on this information, you can evaluate technical and economic feasibility.
- 2. Design the requirements:** When you have identified the project, work with stakeholders to define requirements. You can use the user flow diagram or the high-level UML diagram to show the work of new features and show how it will apply to your existing system.
- 3. Construction/ iteration:** When the team defines the requirements, the work begins. Designers and developers start working on their project, which aims to deploy a working product. The product will undergo various stages of improvement, so it includes simple, minimal functionality.
- 4. Testing:** In this phase, the Quality Assurance team examines the product's performance and looks for the bug.
- 5. Deployment:** In this phase, the team issues a product for the user's work environment.
- 6. Feedback:** After releasing the product, the last step is feedback. In this, the team receives feedback about the product and works through the feedback.

Agile Testing Methods:

- Scrum
- Crystal
- Dynamic Software Development Method(DSDM)
- Feature Driven Development(FDD)
- Lean Software Development
- eXtreme Programming(XP)

Scrum

SCRUM is an agile development process focused primarily on ways to manage tasks in team-based development conditions.

There are three roles in it, and their responsibilities are:

- **Scrum Master:** The scrum can set up the master team, arrange the meeting and remove obstacles for the process
- **Product owner:** The product owner makes the product backlog, prioritizes the delay and is responsible for the distribution of functionality on each repetition.

- **Scrum Team:** The team manages its work and organizes the work to complete the sprint or cycle.

eXtreme Programming(XP)

This type of methodology is used when customers are constantly changing demands or requirements, or when they are not sure about the system's performance.

Crystal

There are three concepts of this method-

1. Chartering: Multi activities are involved in this phase such as making a development team, performing feasibility analysis, developing plans, etc.
2. Cyclic delivery: under this, two more cycles consist, these are:
3. Team updates the release plan.
4. Integrated product delivers to the users.
5. Wrap up: According to the user environment, this phase performs deployment, post-deployment.

Dynamic Software Development Method(DSDM):

DSDM is a rapid application development strategy for software development and gives an agile project distribution structure. The essential features of DSDM are that users must be actively connected, and teams have been given the right to make decisions. The techniques used in DSDM are:

1. Time Boxing
2. MoSCoW Rules
3. Prototyping

The DSDM project contains seven stages:

1. Pre-project
2. Feasibility Study
3. Business Study
4. Functional Model Iteration
5. Design and build Iteration
6. Implementation
7. Post-project

Feature Driven Development(FDD):

This method focuses on "Designing and Building" features. In contrast to other smart methods, FDD describes the small steps of the work that should be obtained separately per function.

Lean Software Development:

Lean software development methodology follows the principle "just in time production." The lean method indicates the increasing speed of software development and reducing costs. Lean development can be summarized in seven phases.

1. Eliminating Waste
2. Amplifying learning
3. Defer commitment (deciding as late as possible)
4. Early delivery
5. Empowering the team
6. Building Integrity
7. Optimize the whole

When to use the Agile Model?

- When frequent changes are required.
- When a highly qualified and experienced team is available.
- When a customer is ready to have a meeting with a software team all the time.
- When project size is small.

Advantage(Pros) of Agile Method:

- Frequent Delivery
- Face-to-Face Communication with clients.
- Efficient design and fulfils the business requirement.
- Anytime changes are acceptable.
- It reduces total development time.

Disadvantages (Cons) of Agile Model:

- Due to the shortage of formal documents, it creates confusion and crucial decisions taken throughout various phases can be misinterpreted at any time by different team members.
- Due to the lack of proper documentation, once the project completes and the developers allotted to another project, maintenance of the finished project can become a difficulty.

Source: <https://www.javatpoint.com/software-engineering-agile-model>

Iterative Model

In this Model, you can start with some of the software specifications and develop the first version of the software. After the first version if there is a need to change the software, then a new version of the software is created with a new iteration. Every release of the Iterative Model finishes in an exact and fixed period that is called iteration.

The Iterative Model allows the accessing earlier phases, in which the variations made respectively. The final output of the project renewed at the end of the Software Development Life Cycle (SDLC) process.

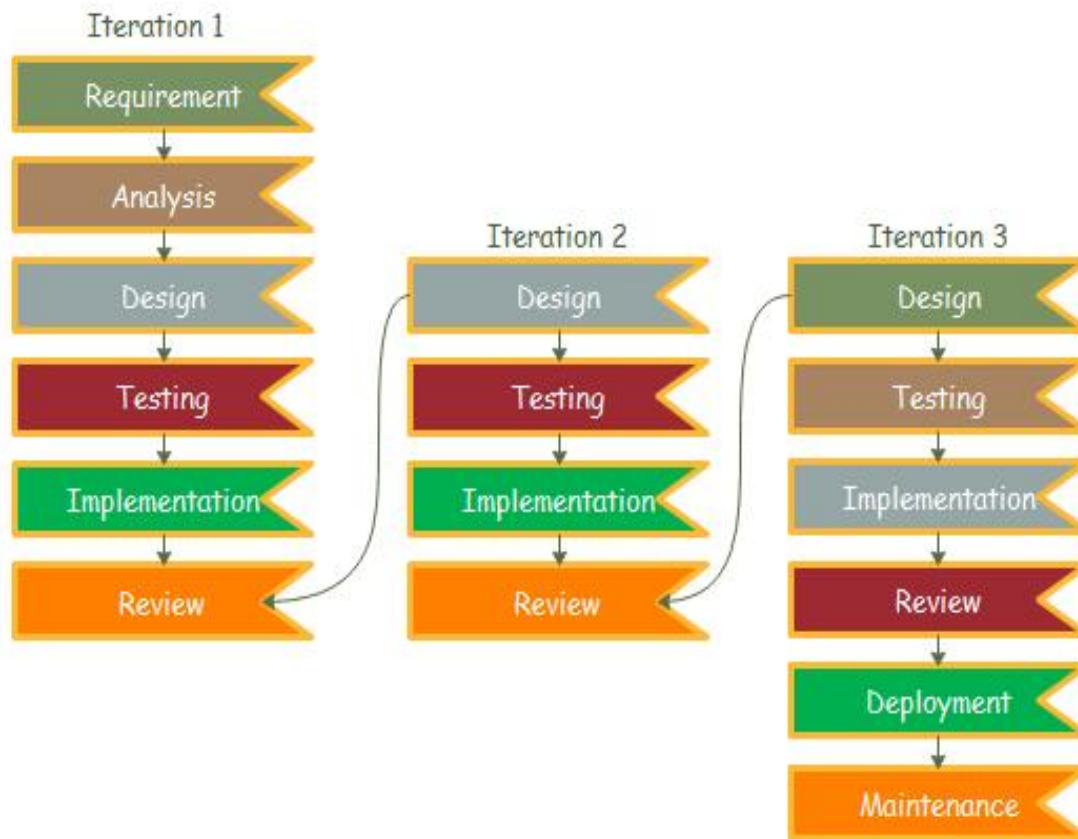


Figure 4.17 Iterative Model

The various phases of Iterative model are as follows:

- 1. Requirement gathering & analysis:** In this phase, requirements are gathered from customers and check by an analyst whether requirements will fulfil or not. Analyst checks that need will achieve within budget or not. After all of this, the software team skips to the next phase.
- 2. Design:** In the design phase, team design the software by the different diagrams like Data Flow diagram, activity diagram, class diagram, state transition diagram, etc.
- 3. Implementation:** In the implementation, requirements are written in the coding language and transformed into computer programs which are called Software.
- 4. Testing:** After completing the coding phase, software testing starts using different test methods. There are many test methods, but the most common are white box, black box, and grey box test methods.
- 5. Deployment:** After completing all the phases, software is deployed to its work environment.
- 6. Review:** In this phase, after the product deployment, review phase is performed to check the behavior and validity of the developed product. And if there are any error found then the process starts again from the requirement gathering.

- 7. Maintenance:** In the maintenance phase, after deployment of the software in the working environment there may be some bugs, some errors or new updates are required. Maintenance involves debugging and new addition options.

When to use the Iterative Model?

- When requirements are defined clearly and easy to understand.
- When the software application is large.
- When there is a requirement of changes in future.

Advantage (Pros) of Iterative Model:

- Testing and debugging during smaller iteration is easy.
- A Parallel development can plan.
- It is easily acceptable to ever-changing needs of the project.
- Risks are identified and resolved during iteration.
- Limited time spent on documentation and extra time on designing.

Disadvantage (Cons) of Iterative Model:

- It is not suitable for smaller projects.
- More Resources may be required.
- Design can be changed again and again because of imperfect requirements.
- Requirement changes can cause over budget.
- Project completion date not confirmed because of changing requirements.

Source: <https://www.javatpoint.com/software-engineering-iterative-model>

Big Bang Model

In this model, developers do not follow any specific process. Development begins with the necessary funds and efforts in the form of inputs. And the result may or may not be as per the customer's requirement, because in this model, even the customer requirements are not defined.

This model is ideal for small projects like academic projects or practical projects. One or two developers can work together on this model.



Figure 4.18 Big Bang Model

When to use Big Bang Model?

As we discussed above, this model is required when this project is small like an academic project or a practical project. This method is also used when the size of the developer team is small and when requirements are not defined, and the release date is not confirmed or given by the customer.

Advantage (Pros) of Big Bang Model:

- There is no planning required.
- Simple Model.
- Few resources required.
- Easy to manage.
- Flexible for developers.

Disadvantage (Cons) of Big Bang Model:

- There are high risk and uncertainty.
- Not acceptable for a large project.
- If requirements are not clear that can cause very expensive.

Source: <https://www.javatpoint.com/software-engineering-big-bang-model>

Prototype Model

The prototype model requires that before carrying out the development of actual software, a working prototype of the system should be built. A prototype is a toy implementation of the system. A prototype usually turns out to be a very crude version of the actual system, possibly exhibiting limited functional capabilities, low reliability, and inefficient performance as compared to actual software. In many instances, the client only has

a general view of what is expected from the software product. In such a scenario where there is an absence of detailed information regarding the input to the system, the processing needs, and the output requirement, the prototyping model may be employed.

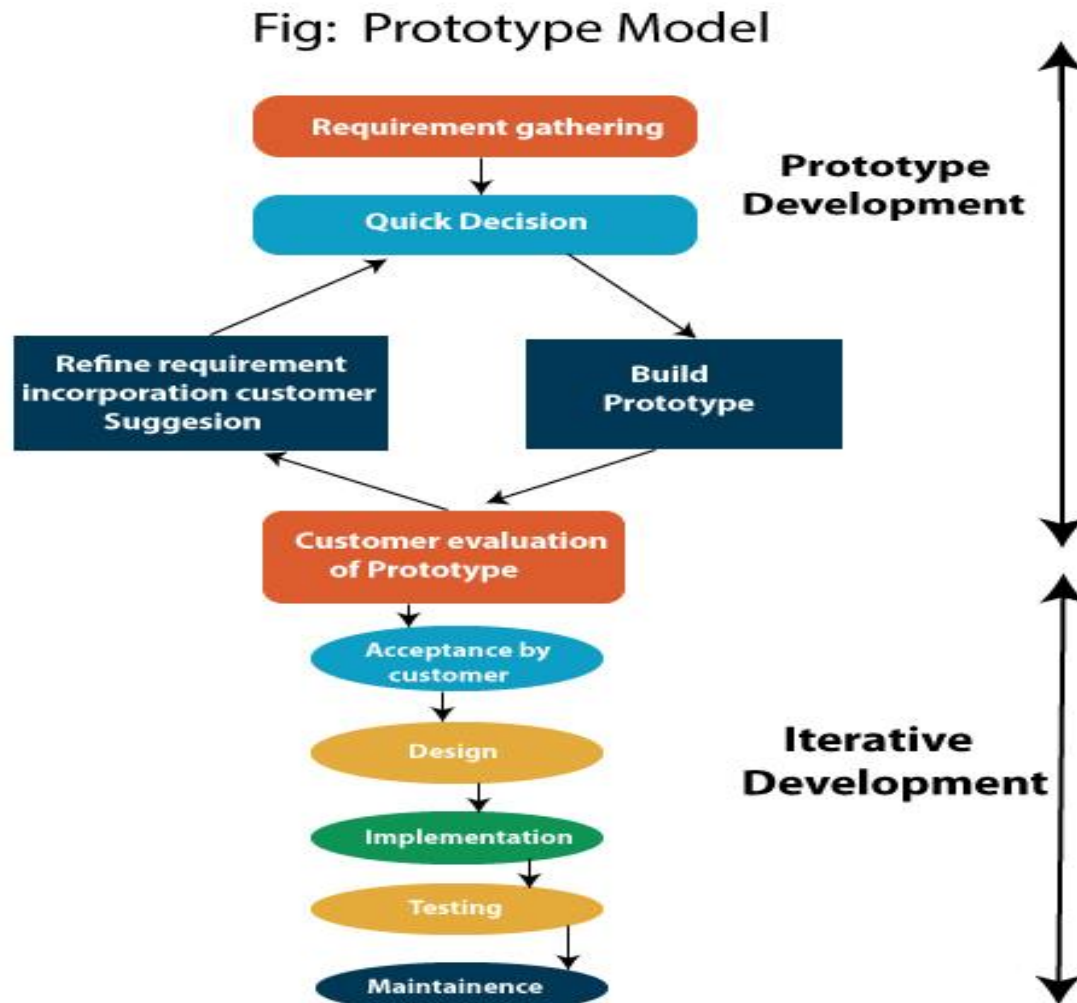


Figure 4.19 Prototype Model

Steps of Prototype Model

- ✓ Requirement Gathering and Analyst
- ✓ Quick Decision
- ✓ Build a Prototype
- ✓ Assessment or User Evaluation
- ✓ Prototype Refinement
- ✓ Engineer Product

Advantage of Prototype Model

- Reduce the risk of incorrect user requirement
- Good where requirement are changing/uncommitted
- Regular visible process aids management
- Support early product marketing
- Reduce Maintenance cost.

- Errors can be detected much earlier as the system is made side by side.

Disadvantage of Prototype Model

- An unstable/badly implemented prototype often becomes the final product.
- Require extensive customer collaboration
- Costs customer money
- Needs committed customer
- Difficult to finish if customer withdraw
- May be too customer specific, no broad market
- Difficult to know how long the project will last.
- Easy to fall back into the code and fix without proper requirement analysis, design, customer evaluation, and feedback.
- Prototyping tools are expensive.
- Special tools & techniques are required to build a prototype.
- It is a time-consuming process.

Evolutionary Process Model

Evolutionary process model resembles the iterative enhancement model. The same phases are defined for the waterfall model occurs here in a cyclical fashion. This model differs from the iterative enhancement model in the sense that this does not require a useful product at the end of each cycle. In evolutionary development, requirements are implemented by category rather than by priority.

For example, in a simple database application, one cycle might implement the graphical user Interface (GUI), another file manipulation, another queries and another updates. All four cycles must complete before there is a working product available. GUI allows the users to interact with the system, file manipulation allow the data to be saved and retrieved, queries allow user to get out of the system, and updates allows users to put data into the system.

Benefits of Evolutionary Process Model

- ❖ Use of EVO brings a significant reduction in risk for software projects.
- ❖ EVO can reduce costs by providing a structured, disciplined avenue for experimentation.
- ❖ EVO allows the marketing department access to early deliveries, facilitating the development of documentation and demonstration.
- ❖ Better fit the product to user needs and market requirements.
- ❖ Manage project risk with the definition of early cycle content.
- ❖ Uncover key issues early and focus attention appropriately.
- ❖ Increase the opportunity to hit market windows.
- ❖ Accelerate sales cycles with early customer exposure.
- ❖ Increase management visibility of project progress.
- ❖ Increase product team productivity and motivations.

Source: <https://www.javatpoint.com/software-engineering-prototype-model>

4.4 Design of Software, Systems, Product and or Processes

The discussion in this phase emphasis on how the proposed system will be designed in accordance with the standard. It focusses on how the requirement analysis and

documentation have been complied with. Especially, those possible tradeoffs presented within the given requirements under various constraints.

The conceptual design, IPO diagrams, data models, systems architecture, and block diagrams are required to be discussed. This will show that the proponents have an idea on how to identify, formulate and solve computing problems.

Software Design

Software design is a mechanism to transform user requirements into some suitable form, which helps the programmer in software coding and implementation. It deals with representing the client's requirement, as described in SRS (Software Requirement Specification) document, into a form, i.e., easily implementable using programming language.

The software design phase is the first step in **SDLC (Software Design Life Cycle)**, which moves the concentration from the problem domain to the solution domain. In software design, we consider the system to be a set of components or modules with clearly defined behaviors & boundaries.

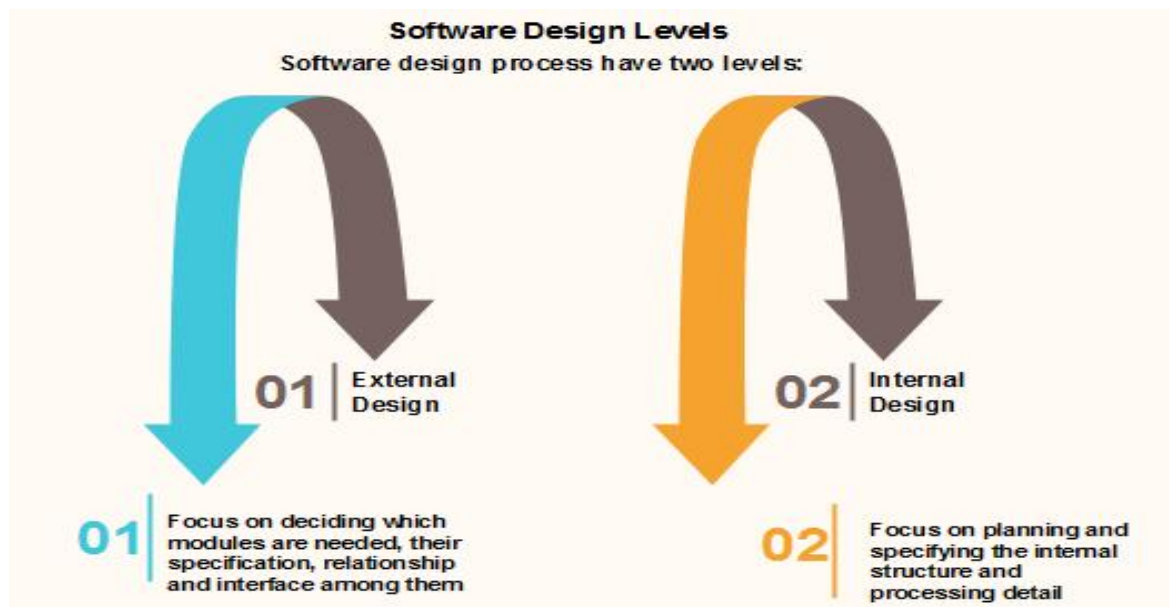


Figure 4.20 Software Design Levels



Figure 4.21 Objectives of Software Design

Objectives of Software Design

Following are the purposes of Software design:

1. **Correctness:** Software design should be correct as per requirement.
2. **Completeness:** The design should have all components like data structures, modules, and external interfaces, etc.
3. **Efficiency:** Resources should be used efficiently by the program.
4. **Flexibility:** Able to modify on changing needs.
5. **Consistency:** There should not be any inconsistency in the design.
6. **Maintainability:** The design should be so simple so that it can be easily maintainable by other designers.

Source: <https://www.javatpoint.com/software-engineering-software-design>

4.5 Implementation Plan

The Implementation Plan describes how the information system will be deployed, installed, and transitioned into an operational system. The plan contains an overview of the system, a brief description of the major tasks involved in the implementation, the overall resources needed to support the implementation effort (such as hardware, software, facilities, materials, and personnel), and any site-specific implementation requirements (Lee, 2013).

Example:

STRATEGY	ACTIVITIES	PERSONS INVOLVED	DURATION
Approval from the user/administrator/ head of office/owner of the business	Letters for the user/administrator/ head of office/ owner of the business	Researchers/ user/administrator/ head of office/ owner of the business	1 Day
System's Installation	Installation of the system and required software and hardware	Researchers/ user/administrator/ head of office/ owner of the business	5 hours
Information Distribution	FLYERS	user/administrator/ head of office/ owner of the business	1 day
	POSTERS	user/administrator/ head of office/ owner of the business	
	MANUALS	user/administrator/ head of office/ owner of the business	
3 Day Training	Hands on Training and Lectures	Researchers/ user/ administrator/ head of office/ owner of the business	3 days

Table 4.1 Sample Implementation Plan

Name:	Date:
--------------	--------------

● Review Questions

1. What is the meaning of methodology?

[illegible]

2. Why requirement analysis is important in the proposed capstone project?

[illegible]

5. Discuss the requirement documentation.

[illegible]

● Assessment Task

-
- This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on its right side, suggesting it's resting on a surface.

6. Give at least two (2) foreign and two (2) latest related literature to your approved title, identify the areas of controversy or challenge and explain how your approved capstone project title will respond to the challenge and in the last paragraph how it is related to your study. Provide additional paper for your answer. It will be strictly hand written.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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